

Junior Engineer Civil Mechanical and Electrical Examination 2024 Paper I

Roll Number	
Candidate Name	
Venue Name	
Exam Date	05/06/2024
Exam Time	1:00 PM - 3:00 PM
Subject	Junior Engineer 2024 Mechanical Paper I

Section : General Intelligence and Reasoning

Q.1 A, B, C, D, E and F are sitting around a circular table facing the centre (not necessarily in the same order). B is sitting to the immediate right of F and immediate left of C. C is sitting to the immediate left of E. A is sitting to the immediate right of E and to the immediate left of D. Who is sitting to the immediate left of F?

- Ans
- ☒ 1. A
 - ☒ 2. C
 - ☒ 3. D
 - ☒ 4. B

Question ID : 630680377010
Option 1 ID : 6306801469105
Option 2 ID : 6306801469104
Option 3 ID : 6306801469106
Option 4 ID : 6306801469103
Status : Answered
Chosen Option : 3

Q.2 What should come in place of '?' in the given series?
394, 465, 536, 607, 678, ?

- Ans
- ☒ 1. 768
 - ☒ 2. 736
 - ☒ 3. 756
 - ☒ 4. 749

Question ID : 630680378803
Option 1 ID : 6306801476262
Option 2 ID : 6306801476260
Option 3 ID : 6306801476259
Option 4 ID : 6306801476261
Status : Answered
Chosen Option : 4

Q.3 What should come in place of the question mark (?) in the given series based on the English alphabetical order?
DMY, BKW, ZIU, XGS, ?

- Ans
- ☒ 1. UDP
 - ☒ 2. VEQ
 - ☒ 3. WCR
 - ☒ 4. UFR

Question ID : 630680382493
Option 1 ID : 6306801490889
Option 2 ID : 6306801490890
Option 3 ID : 6306801490887
Option 4 ID : 6306801490888
Status : Answered
Chosen Option : 2

Q.4 This question consists of a pair of words which have a certain relationship to each other. Select the pair which has the same relationship.

(The words must be considered as meaningful English words and must not be related to each other based on the number of letters/number of consonants/vowels in the word.)

Altimeter : Altitudes

- Ans ☒ 1. Protractor : Light
☒ 2. Taseometer : Wind
☒ 3. Hygrometer : Humidity
☒ 4. Fathometer : Presure

Question ID : 630680556096
Option 1 ID : 6306802173762
Option 2 ID : 6306802173763
Option 3 ID : 6306802173761
Option 4 ID : 6306802173760
Status : Answered
Chosen Option : 3

Q.5 IMPU is related to LPSX in a certain way based on the English alphabetical order. In the same way, BEKN is related to EHNQ. To which of the following is CLQS related, following the same logic?

- Ans ☒ 1. IJSH
☒ 2. ACLK
☒ 3. PLNF
☒ 4. FOTV

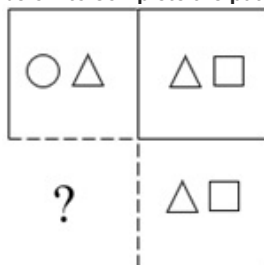
Question ID : 630680377696
Option 1 ID : 6306801471843
Option 2 ID : 6306801471846
Option 3 ID : 6306801471845
Option 4 ID : 6306801471844
Status : Answered
Chosen Option : 4

Q.6 Six friends Abby, Bunny, Chan, Dolly, Emma and Fanny have different weights. Dolly's weight is an odd number. Dolly is heavier than Emma but not the heaviest. Chan is heavier than Fanny but lighter than Dolly. Chan is not heavier than Emma but is heavier than Fanny and Abby. Abby's weight is not an odd number. The lightest weight is 45 kilograms and the heaviest weight is 80 kilograms. Who is the heaviest person in the group?

- Ans ☒ 1. Abby
☒ 2. Emma
☒ 3. Chan
☒ 4. Bunny

Question ID : 630680521125
Option 1 ID : 6306802036925
Option 2 ID : 6306802036928
Option 3 ID : 6306802036926
Option 4 ID : 6306802036927
Status : Answered
Chosen Option : 4

Q.7 Select the option figure that can replace the question mark (?) in the figure given below to complete the pattern.



Ans

✗ 1.



✗ 2.



✗ 3.



✓ 4.



Question ID : 630680525486
 Option 1 ID : 6306802054031
 Option 2 ID : 6306802054028
 Option 3 ID : 6306802054030
 Option 4 ID : 6306802054029

Status : Answered

Chosen Option : 4

Q.8 In a certain code language, 'apple is healthy' is written as 'di jl ew' and 'apple is red' is written as 'di ko ew'. How is 'red' written in the given language?

Ans ✗ 1. ew

✓ 2. ko

✗ 3. jl

✗ 4. di

Question ID : 630680388165
 Option 1 ID : 6306801513224
 Option 2 ID : 6306801513223
 Option 3 ID : 6306801513225
 Option 4 ID : 6306801513222

Status : Answered

Chosen Option : 2

Q.9 LQNR is related to PURV in a certain way based on the English alphabetical order. In the same way, OTQU is related to SXUY. To which of the following is GLIM related, following the same logic?

- Ans
- ☒ 1. QPMK
 - ☒ 2. PQMK
 - ☒ 3. PKMQ
 - ☒ 4. KPMQ

Question ID : 630680379505
Option 1 ID : 6306801479066
Option 2 ID : 6306801479063
Option 3 ID : 6306801479064
Option 4 ID : 6306801479065
Status : Answered
Chosen Option : 4

Q.10 142 is related to 92 following a certain logic. Following the same logic, 255 is related to 205. To which of the following is 457 related following the same logic?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

- Ans
- ☒ 1. 407
 - ☒ 2. 417
 - ☒ 3. 427
 - ☒ 4. 437

Question ID : 630680396913
Option 1 ID : 6306801547720
Option 2 ID : 6306801547721
Option 3 ID : 6306801547722
Option 4 ID : 6306801547723
Status : Answered
Chosen Option : 1

Q.11 15 is related to 230 following a certain logic. Following the same logic, 4 is related to 65. To which of the following is 12 related following the same logic?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting /multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

- Ans
- ☒ 1. 187
 - ☒ 2. 186
 - ☒ 3. 185
 - ☒ 4. 180

Question ID : 630680451715
Option 1 ID : 6306801763946
Option 2 ID : 6306801763947
Option 3 ID : 6306801763948
Option 4 ID : 6306801763949
Status : Answered
Chosen Option : 2

Q.12 Select the set in which the numbers are related in the same way as are the numbers of the following sets.

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(9, 576, 8)

(7, 252, 6)

Ans ☒ 1. (8, 650, 10)

☒ 2. (10, 40, 2)

☒ 3. (4, 82, 5)

☒ 4. (15, 210, 7)

Question ID : 630680390232

Option 1 ID : 6306801521372

Option 2 ID : 6306801521370

Option 3 ID : 6306801521373

Option 4 ID : 6306801521371

Status : Not Answered

Chosen Option : --

Q.13 What will come in place of the question mark (?) in the following equation if '+' and '÷' are interchanged and 'x' and '-' are interchanged?

$$21 \div 15 \times 52 + 13 - 7 = ?$$

Ans ☒ 1. 12

☒ 2. 10

☒ 3. 7

☒ 4. 8

Question ID : 630680376976

Option 1 ID : 6306801468968

Option 2 ID : 6306801468970

Option 3 ID : 6306801468967

Option 4 ID : 6306801468969

Status : Answered

Chosen Option : 4

Q.14 STAR is related to TWCV in a certain way based on the English alphabetical order. In the same way, PARK is related to QDTO. To which of the following is MILK related, following the same logic?

Ans ☒ 1. NLMO

☒ 2. MLNO

☒ 3. MLMO

☒ 4. NLNO

Question ID : 630680376880

Option 1 ID : 6306801468586

Option 2 ID : 6306801468584

Option 3 ID : 6306801468583

Option 4 ID : 6306801468585

Status : Answered

Chosen Option : 4

Q.15 Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statements.

Statements: Some cars are grass. All trucks are cars.

Conclusion 1: Some trucks are not cars.

Conclusion 2: Some trucks are grass.

- Ans**
- ☒ 1. Only conclusion 2 follows
 - ☒ 2. Only conclusion 1 follows
 - ☒ 3. Neither conclusion 1 nor 2 follows
 - ☒ 4. Both conclusion 1 and 2 follow

Question ID : 630680381042
Option 1 ID : 6306801485144
Option 2 ID : 6306801485143
Option 3 ID : 6306801485145
Option 4 ID : 6306801485146
Status : Answered
Chosen Option : 3

Q.16 OB 8 is related to UE 3 in a certain way. In the same way, TO 11 is related to ZR 6. To which of the following is IB 8 related, following the same logic?

- Ans**
- ☒ 1. OF 3
 - ☒ 2. PF -3
 - ☒ 3. PE 3
 - ☒ 4. OE 3

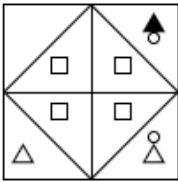
Question ID : 630680386527
Option 1 ID : 6306801506819
Option 2 ID : 6306801506821
Option 3 ID : 6306801506820
Option 4 ID : 6306801506818
Status : Answered
Chosen Option : 4

Q.17 Select the option figure in which the given figure is embedded as its part (rotation is NOT allowed).

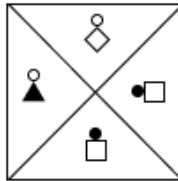


Ans

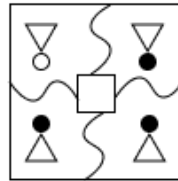
✗ 1.



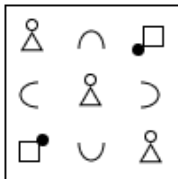
✗ 2.



✓ 3.



✗ 4.



Question ID : 630680422243
Option 1 ID : 6306801647408
Option 2 ID : 6306801647409
Option 3 ID : 6306801647410
Option 4 ID : 6306801647411
Status : Answered
Chosen Option : 3

Q.18 Select the option in which the numbers share the same relationship as that shared by the given number triads.

120-100-60

240-220-180

(NOTE: Operations should be performed on the whole number, without breaking down the numbers into its constituent digits. E.g. 13- Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is NOT allowed.)

Ans ✗ 1. 90-80-75

✗ 2. 150-160-140

✓ 3. 200-180-140

✗ 4. 220-210-200

Question ID : 630680387173
Option 1 ID : 6306801509389
Option 2 ID : 6306801509388
Option 3 ID : 6306801509386
Option 4 ID : 6306801509387
Status : Answered
Chosen Option : 3

Q.19 Select the set in which the numbers are related in the same way as are the numbers of the following sets.

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(45, 23, 73)

(19, 27, 51)

Ans ☒ 1. (22, 29, 52)

☒ 2. (31, 17, 48)

☒ 3. (16, 14, 35)

☒ 4. (12, 19, 30)

Question ID : 630680376817

Option 1 ID : 6306801468332

Option 2 ID : 6306801468333

Option 3 ID : 6306801468334

Option 4 ID : 6306801468331

Status : Answered

Chosen Option : 3

Q.20 What should come in place of the question mark (?) in the given series based on the

English alphabetical order?

RQO, LKI, FEC, ZYW, TSQ, ?

Ans ☒ 1. NMK

☒ 2. NML

☒ 3. MNL

☒ 4. MNK

Question ID : 630680395304

Option 1 ID : 6306801541423

Option 2 ID : 6306801541420

Option 3 ID : 6306801541422

Option 4 ID : 6306801541421

Status : Answered

Chosen Option : 1

Q.21 In a certain code language,

'A + B' means 'A is the brother of B',

'A - B' means 'A is the mother of B',

'A × B' means 'A is the wife of B' and

'A ÷ B' means 'A is the father of B'.

How is B related to L if 'B × G ÷ T + K - L'?

Ans ☒ 1. Father's father

☒ 2. Mother's mother

☒ 3. Sister's daughter

☒ 4. Brother's daughter

Question ID : 630680377669

Option 1 ID : 6306801471736

Option 2 ID : 6306801471735

Option 3 ID : 6306801471738

Option 4 ID : 6306801471737

Status : Answered

Chosen Option : 2

Q.22 The position(s) of how many letters will remain unchanged if each of the letters in the word FOREIGN is arranged in the English alphabetical order?

Ans ☒ 1. None

☐ 2. One

☐ 3. Two

☐ 4. Three

Question ID : 630680404425

Option 1 ID : 6306801577200

Option 2 ID : 6306801577201

Option 3 ID : 6306801577202

Option 4 ID : 6306801577203

Status : Answered

Chosen Option : 1

Q.23 Select the correct option that indicates the arrangement of the following planets according to their size.

1. Mars

2. Jupiter

3. Venus

4. Earth

5. Mercury

Ans ☐ 1. 5, 3, 4, 2, 1

☐ 2. 5, 2, 1, 3, 4

☐ 3. 5, 4, 2, 1, 3

☒ 4. 5, 1, 3, 4, 2

Question ID : 630680565832

Option 1 ID : 6306802212278

Option 2 ID : 6306802212276

Option 3 ID : 6306802212277

Option 4 ID : 6306802212279

Status : Answered

Chosen Option : 4

Q.24 Select the correct mirror image of the given figure when the mirror is placed at MN as shown below.



- Ans
- ☒ 1. F 4 ¥ X R
 - ☒ 2. R X ¥ 4 F
 - ☒ 3. R X ¥ 4 F
 - ☒ 4. F 4 ¥ X R

Question ID : 630680456654
 Option 1 ID : 6306801783579
 Option 2 ID : 6306801783580
 Option 3 ID : 6306801783581
 Option 4 ID : 6306801783582
 Status : Answered
 Chosen Option : 3

Q.25 What will come in the place of the question mark (?) in the following equation, if '+' and 'x' are interchanged and '-' and '÷' are interchanged?
 $12 \div 4 - 24 \times 11 \div 13 = ?$

- Ans
- ☒ 1. -20
 - ☒ 2. 10
 - ☒ 3. 0
 - ☒ 4. -10

Question ID : 630680377614
 Option 1 ID : 6306801471515
 Option 2 ID : 6306801471518
 Option 3 ID : 6306801471517
 Option 4 ID : 6306801471516
 Status : Answered
 Chosen Option : 3

Q.26 What should come in place of the question mark (?) in the given series based on the English alphabetical order?
 INJ, LQM, OTP, RWS, ?

- Ans
- ☒ 1. VUZ
 - ☒ 2. UVZ
 - ☒ 3. UZV
 - ☒ 4. VZU

Question ID : 630680376923
 Option 1 ID : 6306801468757
 Option 2 ID : 6306801468756
 Option 3 ID : 6306801468755
 Option 4 ID : 6306801468758
 Status : Answered
 Chosen Option : 3

Q.27 The position(s) of how many letters will remain unchanged if each of the letters in the word STAMPED is arranged in the English alphabetical order?

- Ans ☒ 1. Two
☐ 2. Three
☐ 3. Four
☐ 4. One

Question ID : 630680396640
Option 1 ID : 6306801546652
Option 2 ID : 6306801546653
Option 3 ID : 6306801546654
Option 4 ID : 6306801546655
Status : Answered
Chosen Option : 1

Q.28 What should come in place of the question mark (?) in the given series based on the English alphabetical order?
LXC, OZD, RBE, UDF, ?

- Ans ☒ 1. XFG
☐ 2. YGH
☐ 3. ZDI
☐ 4. XHJ

Question ID : 630680386936
Option 1 ID : 6306801508454
Option 2 ID : 6306801508455
Option 3 ID : 6306801508456
Option 4 ID : 6306801508457
Status : Answered
Chosen Option : 1

Q.29 In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets.
(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g., 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)
(6, 34)
(8, 46)

- Ans ☐ 1. (4, 24)
☐ 2. (12, 76)
☒ 3. (10, 58)
☐ 4. (7, 38)

Question ID : 630680377730
Option 1 ID : 6306801471982
Option 2 ID : 6306801471980
Option 3 ID : 6306801471979
Option 4 ID : 6306801471981
Status : Answered
Chosen Option : 3

Q.30 In a certain code language, 'HAIR' is coded as '7935' and 'AGED' is coded as '4892'.
What is the code for 'A' in the given language?

Ans  1. 9

 2. 5

 3. 4

 4. 3

Question ID : **630680376549**

Option 1 ID : **6306801467265**

Option 2 ID : **6306801467264**

Option 3 ID : **6306801467266**

Option 4 ID : **6306801467263**

Status : **Answered**

Chosen Option : **1**

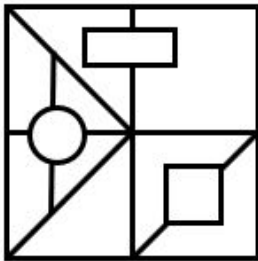
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Q.31 Select the option in which the given figure is embedded (rotation is NOT allowed).

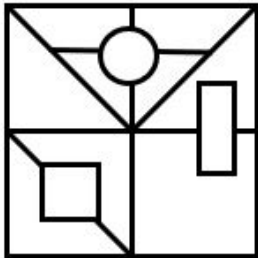


Ans

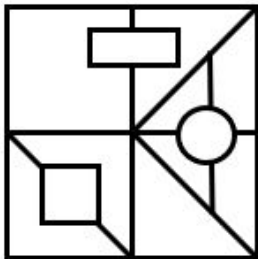
✓ 1.



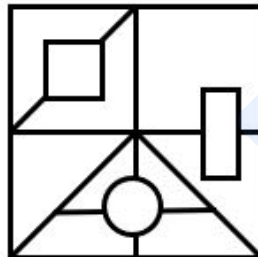
✗ 2.



✗ 3.



✗ 4.



Question ID : 630680524478

Option 1 ID : 6306802050014

Option 2 ID : 6306802050011

Option 3 ID : 6306802050013

Option 4 ID : 6306802050012

Status : Answered

Chosen Option : 1

Q.32 What will come in the place of the question mark (?) in the following equation, if '+' and '÷' are interchanged and 'x' and '-' are interchanged?
 $10 + 2 - 10 \div 5 \times 10 = ?$

- Ans ☒ 1. 40
☒ 2. 45
☒ 3. 35
☒ 4. 32

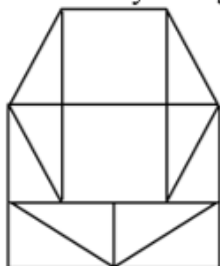
Question ID : 630680377708
 Option 1 ID : 6306801471892
 Option 2 ID : 6306801471893
 Option 3 ID : 6306801471894
 Option 4 ID : 6306801471891
 Status : Answered
 Chosen Option : 2

Q.33 What should come in place of the question mark (?) in the given series.
 6, 3, 12, 6, 24, ?, 48, 24, 96

- Ans ☒ 1. 6
☒ 2. 12
☒ 3. 24
☒ 4. 48

Question ID : 630680379055
 Option 1 ID : 6306801477267
 Option 2 ID : 6306801477269
 Option 3 ID : 6306801477268
 Option 4 ID : 6306801477270
 Status : Answered
 Chosen Option : 2

Q.34 How many triangles are there in the given figure?



- Ans ☒ 1. 11
☒ 2. 13
☒ 3. 12
☒ 4. 14

Question ID : 630680524413
 Option 1 ID : 6306802049751
 Option 2 ID : 6306802049753
 Option 3 ID : 6306802049752
 Option 4 ID : 6306802049754
 Status : Answered
 Chosen Option : 2

Q.35 Select the option that indicates the arrangement of the following words in meaningful and logical order.

1. Word
2. Paragraph
3. Sentence
4. Letter
5. Phrase

Ans ☒ 1. 1, 3, 2, 4, 5
☒ 2. 2, 1, 5, 4, 3
☒ 3. 5, 1, 3, 2, 4
☒ 4. 4, 1, 5, 3, 2

Question ID : 630680404456
 Option 1 ID : 6306801577324
 Option 2 ID : 6306801577325
 Option 3 ID : 6306801577326
 Option 4 ID : 6306801577327
 Status : Answered

Chosen Option : 1

Q.36 A, B, C, D, E, and F are sitting at a circular table facing the centre (not necessarily in the same order). A is to the immediate left of B, E is third to the right of B, D is to the immediate right of C. F is third to the right of C.

Who are the immediate neighbours of E?

- a) A and B
- b) B and D
- c) D and F
- d) F and B

Ans ☒ 1. (d)
☒ 2. (c)
☒ 3. (a)
☒ 4. (b)

Question ID : 630680376401
 Option 1 ID : 6306801466674
 Option 2 ID : 6306801466673
 Option 3 ID : 6306801466671
 Option 4 ID : 6306801466672
 Status : Answered

Chosen Option : 2

Q.37 Select the word-pair that best represents a similar relationship to the one expressed in the pair of words given below.

(The words must be considered as meaningful English words and must not be related to each other based on the number of letters/number of consonants/vowels in the word.)

Mycology : Fungi

Ans ☒ 1. Ornithology : Eggs
☒ 2. Entomology : Birds
☒ 3. Seismology : Earthquakes
☒ 4. Paedology : Planets

Question ID : 630680556091
 Option 1 ID : 6306802173742
 Option 2 ID : 6306802173741
 Option 3 ID : 6306802173740
 Option 4 ID : 6306802173743
 Status : Answered

Chosen Option : 3

Q.38 What should come in place of the question mark (?) in the given series based on the English alphabetical order
KMO, HJL, EGI, BDF, ?

- Ans
- ☒ 1. ZAC
 - ☒ 2. YBC
 - ☒ 3. ZBC
 - ☒ 4. YAC

Question ID : 630680376901
Option 1 ID : 6306801468670
Option 2 ID : 6306801468668
Option 3 ID : 6306801468667
Option 4 ID : 6306801468669
Status : Answered
Chosen Option : 4

Q.39 In a certain code language,
A + B means 'A is the sister of B'
A # B means 'A is the brother of B'
A × B means 'A is the wife of B'
A @ B means 'A is the father of B'
Based on the above, how is E related to M if 'E # F × H @ G + M'?

- Ans
- ☒ 1. Mother's brother
 - ☒ 2. Mother's father
 - ☒ 3. Brother
 - ☒ 4. Father

Question ID : 630680376959
Option 1 ID : 6306801468900
Option 2 ID : 6306801468901
Option 3 ID : 6306801468902
Option 4 ID : 6306801468899
Status : Answered
Chosen Option : 1

Q.40 Madan starts from point A and drives 4 km towards the South. He then takes a left turn and drives 5 km. he then takes a right turn and drives 2 km. He then takes another right turn and drives 3 km. He takes a final right turn and drives 6 km to reach point B. How far (shortest distance) and towards which direction should he drive to reach Point A again? (All turns are 90 degrees turns only unless specified)

- Ans
- ☒ 1. 2 km West
 - ☒ 2. 3 km West
 - ☒ 3. 3 km East
 - ☒ 4. 2 km East

Question ID : 630680376853
Option 1 ID : 6306801468475
Option 2 ID : 6306801468477
Option 3 ID : 6306801468478
Option 4 ID : 6306801468476
Status : Answered
Chosen Option : 1

Q.41 LQKN is related to PUOR in a certain way based on the English alphabetical order. In the same way, INHK is related to MRLO. To which of the following is QVPS related, following the same logic?

- Ans ☐ 1. WZTU
☒ 2. UZTW
☐ 3. WZUT
☐ 4. UZWT

Question ID : 630680376720
Option 1 ID : 6306801467943
Option 2 ID : 6306801467946
Option 3 ID : 6306801467944
Option 4 ID : 6306801467945
Status : Answered
Chosen Option : 2

Q.42 Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements.

Statements: All frames are rocks. All rocks are clips. All clips are towels.

Conclusion (I): All frames are towels.

Conclusion (II): At least some clips are frames.

- Ans ☐ 1. Only conclusion (II) follows
☐ 2. Neither conclusion (I) nor (II) follows
☒ 3. Both conclusions (I) and (II) follow
☐ 4. Only conclusion (I) follows

Question ID : 630680376527
Option 1 ID : 6306801467176
Option 2 ID : 6306801467177
Option 3 ID : 6306801467178
Option 4 ID : 6306801467175
Status : Answered
Chosen Option : 3

Q.43 What should come in place of the question mark (?) in the given series based on the English alphabetical order?

MAT MET MIT MOT ?

- Ans ☐ 1. MVT
☐ 2. MNT
☐ 3. MXT
☒ 4. MUT

Question ID : 630680378698
Option 1 ID : 6306801475839
Option 2 ID : 6306801475840
Option 3 ID : 6306801475841
Option 4 ID : 6306801475842
Status : Answered
Chosen Option : 4

Q.44 What should come in place of the question mark (?) in the given series?
2, 5, 13, 28, 52, 87, ?

- Ans
- ☒ 1. 130
 - ☒ 2. 98
 - ☒ 3. 107
 - ☒ 4. 135

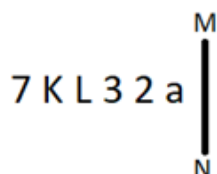
Question ID : 630680378757
Option 1 ID : 6306801476077
Option 2 ID : 6306801476075
Option 3 ID : 6306801476076
Option 4 ID : 6306801476078
Status : Answered
Chosen Option : 4

Q.45 In a certain code language, 'SANG' is coded as '3618', and 'RANG' is coded as '6438'.
What is the code for 'S' in that language?

- Ans
- ☒ 1. 8
 - ☒ 2. 3
 - ☒ 3. 6
 - ☒ 4. 1

Question ID : 630680376410
Option 1 ID : 6306801466710
Option 2 ID : 6306801466707
Option 3 ID : 6306801466708
Option 4 ID : 6306801466709
Status : Answered
Chosen Option : 4

Q.46 Select the correct mirror image of the given figure when the mirror is placed at MN as shown below.



- Ans
- ☒ 1. 6 5 8 J K 7
 - ☒ 2. a 5 L 3 K 7
 - ☒ 3. 2 a 8 J K 7
 - ☒ 4. 5 9 8 J K 7

Question ID : 630680524860
Option 1 ID : 6306802051540
Option 2 ID : 6306802051542
Option 3 ID : 6306802051539
Option 4 ID : 6306802051541
Status : Answered
Chosen Option : 1

Q.47 Jennifer starts from Point D and drives 3 km towards the north. She then takes a left turn and drives 10 km. She then takes another left turn and drives 6 km. She takes one more left turn and drives 5 km. She then takes a right turn and drives 3 km. She takes a final left turn and drives 5 km to reach point E. How far (shortest distance) and towards which direction should she drive in order to reach Point D again? (All turns are 90 degree turns only, unless specified)

- Ans** ☒ 1. 6 km North
☒ 2. 3 km North
☒ 3. 5 km East
☒ 4. 5 km North

Question ID : **630680378906**
 Option 1 ID : **6306801476674**
 Option 2 ID : **6306801476671**
 Option 3 ID : **6306801476673**
 Option 4 ID : **6306801476672**
 Status : **Answered**
 Chosen Option : **1**

Q.48 In a certain code language, 'FLEW' is coded as '8462' and 'LORD' is coded as '9736'. What is the code for 'L' in the given code language?

- Ans** ☒ 1. 9
☒ 2. 7
☒ 3. 6
☒ 4. 4

Question ID : **630680377195**
 Option 1 ID : **6306801469844**
 Option 2 ID : **6306801469846**
 Option 3 ID : **6306801469845**
 Option 4 ID : **6306801469843**
 Status : **Answered**
 Chosen Option : **3**

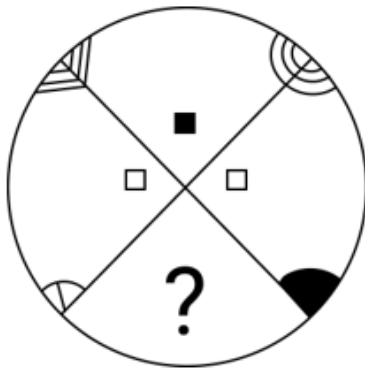
Q.49 If 'A' stands for '+', 'B' stands for 'x', 'C' stands for '+' and 'D' stands for '-', what will come in place of the question mark '?' in the following equation?

$$10 \text{ B } 2 \text{ D } 15 \text{ A } 3 \text{ C } 7 = ?$$

- Ans** ☒ 1. 22
☒ 2. 20
☒ 3. 21
☒ 4. 24

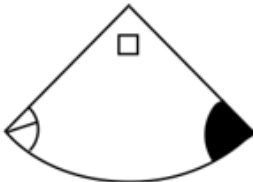
Question ID : **630680502794**
 Option 1 ID : **6306801964757**
 Option 2 ID : **6306801964758**
 Option 3 ID : **6306801964759**
 Option 4 ID : **6306801964760**
 Status : **Answered**
 Chosen Option : **1**

Q.50 Select the figure from the options that can replace the question mark (?) and complete the given pattern.



Ans

✗ 1.



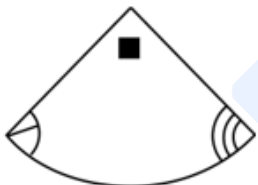
✗ 2.



✓ 3.



✗ 4.



Question ID : 630680524809
Option 1 ID : 6306802051336
Option 2 ID : 6306802051338
Option 3 ID : 6306802051337
Option 4 ID : 6306802051335
Status : Answered
Chosen Option : 3

Section : General Awareness

Q.1 Which Chief Justice of India was sworn in as the acting President of India on 20th July 1969?

- Ans
- ☐ 1. Justice Harilal Jekisundas Kania
 - ☐ 2. Justice M Patanjali Sastri
 - ☐ 3. Justice Sudhi Ranjan Das
 - ☒ 4. Justice Mohammad Hidayatullah

Question ID : 630680407827
Option 1 ID : 6306801590246
Option 2 ID : 6306801590247
Option 3 ID : 6306801590248
Option 4 ID : 6306801590249
Status : Answered
Chosen Option : 4

Q.2 Identify the INCORRECT pair (railway zone and headquarters) from the following.

- Ans
- ☒ 1. Central Railway Zone – Kolkata
 - ☐ 2. North Western Railway Zone – Jaipur
 - ☐ 3. South Western Railway Zone – Hubli
 - ☐ 4. South-East Central Railway Zone – Bilaspur

Question ID : 630680510977
Option 1 ID : 6306801997038
Option 2 ID : 6306801997039
Option 3 ID : 6306801997040
Option 4 ID : 6306801997041
Status : Answered
Chosen Option : 1

Q.3 Which of the following committees was responsible for making recommendations that led to the 42nd Constitutional Amendment for the declaration of Fundamental Duties?

- Ans
- ☒ 1. Swaran Singh Committee
 - ☐ 2. Abid Hussain Committee
 - ☐ 3. Khusro Committee
 - ☐ 4. Raja Chelliah Committee

Question ID : 630680618079
Option 1 ID : 6306802418621
Option 2 ID : 6306802418622
Option 3 ID : 6306802418623
Option 4 ID : 6306802418624
Status : Answered
Chosen Option : 1

Q.4 What is the SI unit of force?

- Ans
- ☒ 1. Newton
 - ☐ 2. Volt
 - ☐ 3. Watt
 - ☐ 4. Joule

Question ID : 630680407778
Option 1 ID : 6306801590053
Option 2 ID : 6306801590052
Option 3 ID : 6306801590051
Option 4 ID : 6306801590050
Status : Answered
Chosen Option : 1

Q.5 When was Swami Vivekananda born?

- Ans**
- ☐ 1. 12 January 1853
 - ☐ 2. 12 January 1859
 - ☒ 3. 12 January 1863
 - ☐ 4. 12 January 1866

Question ID : 630680511596
Option 1 ID : 6306801999506
Option 2 ID : 6306801999507
Option 3 ID : 6306801999508
Option 4 ID : 6306801999509
Status : Answered
Chosen Option : 1

Q.6 It is a fact that if you drink from a plastic bottle and throw it in the dustbin, it will still be there when your grandchildren are old. What is the reason behind this?

- Ans**
- ☐ 1. Plastics are biodegradable.
 - ☐ 2. Plastics are radioactive.
 - ☒ 3. Plastics are non-biodegradable.
 - ☐ 4. Plastics are heavy metals.

Question ID : 630680403334
Option 1 ID : 6306801572845
Option 2 ID : 6306801572846
Option 3 ID : 6306801572847
Option 4 ID : 6306801572848
Status : Answered
Chosen Option : 3

Q.7 Which of the following is the shortcut key to create a new paragraph in MS Word?

- Ans**
- ☐ 1. Ctrl + N
 - ☐ 2. Ctrl + P
 - ☒ 3. Enter
 - ☐ 4. Shift + Enter

Question ID : 630680571752
Option 1 ID : 6306802235916
Option 2 ID : 6306802235917
Option 3 ID : 6306802235918
Option 4 ID : 6306802235919
Status : Answered
Chosen Option : 1

Q.8 During which years did Khalji dynasty ruled over Delhi?

- Ans**
- ☐ 1. 1320 - 1414
 - ☐ 2. 1414 - 1451
 - ☐ 3. 1451 - 1526
 - ☒ 4. 1290 - 1320

Question ID : 630680511634
Option 1 ID : 6306801999659
Option 2 ID : 6306801999660
Option 3 ID : 6306801999661
Option 4 ID : 6306801999658
Status : Answered
Chosen Option : 4

Q.9 Which of the following values is subtracted from the numerical value of temperature expressed in Kelvin to obtain the temperature in degrees Celsius?

- Ans
- ☒ 1. 293
 - ☒ 2. 253
 - ☒ 3. 313
 - ☒ 4. 273

Question ID : **630680403022**
 Option 1 ID : **6306801571661**
 Option 2 ID : **6306801571659**
 Option 3 ID : **6306801571662**
 Option 4 ID : **6306801571660**
 Status : **Answered**
 Chosen Option : **4**

Q.10 Private income minus tax payment minus non-tax payment will give us the estimate for _____.

- Ans
- ☒ 1. private income
 - ☒ 2. national income
 - ☒ 3. gross income
 - ☒ 4. personal disposable income

Question ID : **630680512477**
 Option 1 ID : **6306802002879**
 Option 2 ID : **6306802002881**
 Option 3 ID : **6306802002880**
 Option 4 ID : **6306802002878**
 Status : **Answered**
 Chosen Option : **4**

Q.11 Match the following states with their percentage urban population in India, as per Census 2011.

- i) Kerala a) 10.04%
 ii) Madhya Pradesh b) 47.72%
 iii) Tamil Nadu c) 27.63%
 iv) Himachal Pradesh d) 48.45%

- Ans
- ☒ 1. i) d, ii) c, iii) b, iv) a
 - ☒ 2. i) a, ii) d, iii) c, iv) b
 - ☒ 3. i) a, ii) b, iii) c, iv) d
 - ☒ 4. i) b, ii) c, iii) d, iv) a

Question ID : **630680488637**
 Option 1 ID : **6306801908567**
 Option 2 ID : **6306801908568**
 Option 3 ID : **6306801908569**
 Option 4 ID : **6306801908570**
 Status : **Answered**
 Chosen Option : **1**

Q.12 Which of the following is the correct option for using 'justify' alignment to a paragraph in a document?

- Ans
- ☒ 1. Aligns text only to the left
 - ☒ 2. Aligns text only to the right
 - ☒ 3. Aligns text to both left and right edges
 - ☒ 4. Align text in the center

Question ID : 630680571762
 Option 1 ID : 6306802235952
 Option 2 ID : 6306802235953
 Option 3 ID : 6306802235954
 Option 4 ID : 6306802235955
 Status : Answered
 Chosen Option : 3

Q.13 Which are the two major branches of the southwest monsoon in India?

- Ans
- ☒ 1. The Tibetan plateau branch and the Bay of Bengal branch
 - ☒ 2. The Himalayan Mountain branch and the Gulf of Mannar branch
 - ☒ 3. The Arabian Sea branch and the Tibetan plateau branch
 - ☒ 4. The Arabian Sea branch and the Bay of Bengal branch

Question ID : 630680510958
 Option 1 ID : 6306801996964
 Option 2 ID : 6306801996965
 Option 3 ID : 6306801996963
 Option 4 ID : 6306801996962
 Status : Answered
 Chosen Option : 4

Q.14 What is the primary function of the Golgi apparatus?

- Ans
- ☒ 1. Protein synthesis
 - ☒ 2. Lipid production
 - ☒ 3. Cellular respiration
 - ☒ 4. Sorting and packaging of cellular products

Question ID : 630680540550
 Option 1 ID : 6306802113275
 Option 2 ID : 6306802113276
 Option 3 ID : 6306802113278
 Option 4 ID : 6306802113277
 Status : Answered
 Chosen Option : 4

Q.15 Narmada Bachao Andolan (Save Narmada Movement) is a/an _____.

- Ans
- ☒ 1. forest conservation movement
 - ☒ 2. project of Jal Shakti Ministry
 - ☒ 3. Social movement
 - ☒ 4. government initiative

Question ID : 630680564468
 Option 1 ID : 6306802206805
 Option 2 ID : 6306802206806
 Option 3 ID : 6306802206804
 Option 4 ID : 6306802206803
 Status : Answered
 Chosen Option : 3

Q.16 Who said "a cherry that will drop into our mouth one day" about the kingdom of Awadh?

- Ans
- ☐ 1. Lord Lytton
 - ☐ 2. Lord Curzon
 - ☐ 3. Lord Mountbatten
 - ☒ 4. Lord Dalhousie

Question ID : 630680511542
Option 1 ID : 6306801999292
Option 2 ID : 6306801999290
Option 3 ID : 6306801999293
Option 4 ID : 6306801999291
Status : Answered
Chosen Option : 2

Q.17 Who introduced the Anusandhan National Research Foundation Bill, 2023, in the Lok Sabha?

- Ans
- ☒ 1. Dr. Jitender Singh
 - ☐ 2. Gopal Rai
 - ☐ 3. Imran Hussain
 - ☐ 4. Piyush Goyal

Question ID : 630680469598
Option 1 ID : 6306801834155
Option 2 ID : 6306801834156
Option 3 ID : 6306801834157
Option 4 ID : 6306801834158
Status : Answered
Chosen Option : 1

Q.18 Which of the following bacterial infections often affects the lungs?

- Ans
- ☐ 1. Herpes
 - ☐ 2. Influenza
 - ☒ 3. Tuberculosis
 - ☐ 4. Meningitis

Question ID : 630680404156
Option 1 ID : 6306801576111
Option 2 ID : 6306801576113
Option 3 ID : 6306801576110
Option 4 ID : 6306801576112
Status : Answered
Chosen Option : 2

Q.19 Olympic medalist Vijender Singh belongs to which State of India?

- Ans
- ☐ 1. Punjab
 - ☐ 2. Delhi
 - ☒ 3. Haryana
 - ☐ 4. Maharashtra

Question ID : 630680435739
Option 1 ID : 6306801700873
Option 2 ID : 6306801700874
Option 3 ID : 6306801700875
Option 4 ID : 6306801700876
Status : Answered
Chosen Option : 3

Q.20 Who among the following musicians received the Padma Bhushan Award 2022 for his contributions in the field of art?

- Ans ☒ 1. Chhannulal Mishra
- ☒ 2. Rashid Khan
- ☒ 3. Madan Singh Chauhan
- ☒ 4. Manilal Nag

Question ID : 630680701418
 Option 1 ID : 6306802747611
 Option 2 ID : 6306802747612
 Option 3 ID : 6306802747610
 Option 4 ID : 6306802747609
 Status : Answered
 Chosen Option : 2

Q.21 Who was the leader of the temple entry movement in 1930 at Kalaram temple, Nashik?

- Ans ☒ 1. Swami Achhutanand
- ☒ 2. NG Ranga
- ☒ 3. Mahatma Gandhi
- ☒ 4. Dr. BR Ambedkar

Question ID : 630680572381
 Option 1 ID : 6306802238364
 Option 2 ID : 6306802238365
 Option 3 ID : 6306802238366
 Option 4 ID : 6306802238367
 Status : Answered
 Chosen Option : 4

Q.22 The office of the Registrar General and Census Commissioner works under which Ministry of Government of India?

- Ans ☒ 1. Ministry of Finance
- ☒ 2. Ministry of Personnel, Public Grievances and Pensions
- ☒ 3. Ministry of Home Affairs
- ☒ 4. Ministry of Social Justice and Empowerment

Question ID : 630680477183
 Option 1 ID : 6306801863901
 Option 2 ID : 6306801863902
 Option 3 ID : 6306801863903
 Option 4 ID : 6306801863904
 Status : Answered
 Chosen Option : 2

Q.23 _____ is referred to as paper taxes.

- Ans ☒ 1. Gift tax
- ☒ 2. Corporation tax
- ☒ 3. Customs duty
- ☒ 4. Excise tax

Question ID : 630680402051
 Option 1 ID : 6306801567904
 Option 2 ID : 6306801567905
 Option 3 ID : 6306801567906
 Option 4 ID : 6306801567907
 Status : Answered
 Chosen Option : 1

Q.24 Name the first Finance Minister of independent India, whose statue was inaugurated by the Union Textiles Minister in Coimbatore .

- Ans**
- ☒ 1. RK Shanmugam Chetty
 - ☐ 2. CD Deshmukh
 - ☐ 3. TT Krishnamachari
 - ☐ 4. Sachindra Chaudhuri

Question ID : **630680412664**
Option 1 ID : **6306801609479**
Option 2 ID : **6306801609480**
Option 3 ID : **6306801609481**
Option 4 ID : **6306801609482**
Status : **Answered**
Chosen Option : **1**

Q.25 To cherish and follow the noble ideals that inspired our national struggle for freedom is a _____ fundamental duty in the list of fundamental duties enshrined in our Indian Constitution.

- Ans**
- ☐ 1. first
 - ☒ 2. second
 - ☐ 3. third
 - ☐ 4. fourth

Question ID : **630680493470**
Option 1 ID : **6306801927619**
Option 2 ID : **6306801927620**
Option 3 ID : **6306801927621**
Option 4 ID : **6306801927622**
Status : **Answered**
Chosen Option : **2**

Q.26 If an earthquake happens at the sea floor and one tectonic plate dips under the another, this will most probably lead to a natural disaster known as:

- Ans**
- ☐ 1. Typhoon
 - ☐ 2. Tornado
 - ☒ 3. Tsunami
 - ☐ 4. Cyclone

Question ID : **630680546842**
Option 1 ID : **6306802137938**
Option 2 ID : **6306802137937**
Option 3 ID : **6306802137935**
Option 4 ID : **6306802137936**
Status : **Answered**
Chosen Option : **4**

Q.27 The valency of Ca is 2^+ . The valency of O is 2^- . What is the simplified chemical formula of Calcium oxide?

- Ans
- ☒ 1. Ca_2O_2
 - ☒ 2. CaO
 - ☒ 3. CaO_2
 - ☒ 4. Ca_2O

Question ID : 630680401862
Option 1 ID : 6306801567218
Option 2 ID : 6306801567219
Option 3 ID : 6306801567220
Option 4 ID : 6306801567221
Status : Answered
Chosen Option : 2

Q.28 Who among the following is the author of the book published in December 2021, 'The Monk Who Transformed Uttar Pradesh'?

- Ans
- ☒ 1. Shantanu Gupta
 - ☒ 2. Yashika Dutta
 - ☒ 3. Amish Tripathi
 - ☒ 4. Shashi Tharoor

Question ID : 630680394949
Option 1 ID : 6306801540064
Option 2 ID : 6306801540065
Option 3 ID : 6306801540066
Option 4 ID : 6306801540067
Status : Answered
Chosen Option : 3

Q.29 Which of the following sentences is/are correct?

- i. According to the Census of India 2011, the total population of India is 141 crores.
- ii. Between 1911 and 1921, there was a negative rate of growth of India's population.
- iii. Between 1901 and 1951, the average annual growth of India's population did not exceed 1.33%.

- Ans
- ☒ 1. Only i and ii
 - ☒ 2. Only i and iii
 - ☒ 3. Only i
 - ☒ 4. Only ii and iii

Question ID : 630680488238
Option 1 ID : 6306801907011
Option 2 ID : 6306801907012
Option 3 ID : 6306801907013
Option 4 ID : 6306801907014
Status : Answered
Chosen Option : 4

Q.30 In 1800, which of the following experimental achievements was made by William Nicholson?

- Ans
- ☐ 1. Discovery of thermionic emission
 - ☐ 2. Discovery of low-cost filters for polarizing light
 - ☒ 3. Discovery of water electrolysis
 - ☐ 4. Discovery of electric bulb

Question ID : 630680402063
Option 1 ID : 6306801567953
Option 2 ID : 6306801567952
Option 3 ID : 6306801567951
Option 4 ID : 6306801567950
Status : Answered
Chosen Option : 3

Q.31 Who was elected as the 16th Speaker of the Himachal Pradesh Legislative Assembly in January 2023?

- Ans
- ☐ 1. Suresh Bhardwaj
 - ☐ 2. Jai Ram Thakur
 - ☒ 3. Kuldeep Singh Pathania
 - ☐ 4. Anurag Thakur

Question ID : 630680804277
Option 1 ID : 6306803150842
Option 2 ID : 6306803150839
Option 3 ID : 6306803150840
Option 4 ID : 6306803150841
Status : Answered
Chosen Option : 3

Q.32 Which of the following is a macronutrient?

- Ans
- ☐ 1. Vitamins
 - ☒ 2. Carbohydrates
 - ☐ 3. Minerals
 - ☐ 4. Antioxidants

Question ID : 630680411618
Option 1 ID : 6306801605307
Option 2 ID : 6306801605308
Option 3 ID : 6306801605309
Option 4 ID : 6306801605310
Status : Answered
Chosen Option : 2

Q.33 Arachnids belong to which subphylum of the phylum Arthropoda?

- Ans
- ☐ 1. Crustacea
 - ☒ 2. Chelicerata
 - ☐ 3. Hexapoda
 - ☐ 4. Myriapoda

Question ID : 630680540420
Option 1 ID : 6306802112768
Option 2 ID : 6306802112767
Option 3 ID : 6306802112770
Option 4 ID : 6306802112769
Status : Answered
Chosen Option : 3

Q.34 The government of which of the following states launched the 'Nand Baba Milk Mission' scheme in June 2023?

- Ans
- ☒ 1. Madhya Pradesh
 - ☒ 2. Himachal Pradesh
 - ☒ 3. Uttar Pradesh
 - ☒ 4. Bihar

Question ID : 630680513786
Option 1 ID : 6306802007770
Option 2 ID : 6306802007771
Option 3 ID : 6306802007769
Option 4 ID : 6306802007772
Status : Answered
Chosen Option : 2

Q.35 How do estuaries serve as critical habitats for various species?

- Ans
- ☒ 1. Estuaries have no ecological significance.
 - ☒ 2. Estuaries provide breeding grounds and nurseries for marine organisms.
 - ☒ 3. Estuaries contribute to desertification.
 - ☒ 4. Estuaries are only important for recreational activities.

Question ID : 630680615281
Option 1 ID : 6306802407798
Option 2 ID : 6306802407799
Option 3 ID : 6306802407800
Option 4 ID : 6306802407801
Status : Answered
Chosen Option : 2

Q.36 Which of the following articles of the Constitution of India provides measures to ensure the enforcement of Fundamental Rights?

- Ans
- ☒ 1. Article 29
 - ☒ 2. Article 32
 - ☒ 3. Article 19
 - ☒ 4. Article 14

Question ID : 630680511439
Option 1 ID : 6306801998888
Option 2 ID : 6306801998889
Option 3 ID : 6306801998887
Option 4 ID : 6306801998886
Status : Answered
Chosen Option : 2

Q.37 What is the primary focus of the ASPIRE scheme of the Ministry of Micro, Small & Medium Enterprises, Government of India?

- Ans
- ☒ 1. Urban development
 - ☒ 2. Technology innovation
 - ☒ 3. Environmental conservation
 - ☒ 4. Establishment of livelihoods business incubation centres

Question ID : 630680488098
Option 1 ID : 6306801906451
Option 2 ID : 6306801906452
Option 3 ID : 6306801906453
Option 4 ID : 6306801906454
Status : Answered
Chosen Option : 2

Q.38 Name the scheme launched by the Prime Minister on 17 September 2023 that aims to improve the lives of artisans.

- Ans** ☒ 1. PM Vishwakarma
☐ 2. PM Shilpgram
☐ 3. PM Anantakalpaaya
☐ 4. PM Kalashi

Question ID : **630680411164**
Option 1 ID : **6306801603412**
Option 2 ID : **6306801603413**
Option 3 ID : **6306801603414**
Option 4 ID : **6306801603415**
Status : **Answered**
Chosen Option : **1**

Q.39 Which of the following is a heart-related disease?

- Ans** ☒ 1. Angina
☐ 2. Goiter
☐ 3. Acromegaly
☐ 4. Diabetes

Question ID : **630680544738**
Option 1 ID : **6306802129717**
Option 2 ID : **6306802129716**
Option 3 ID : **6306802129715**
Option 4 ID : **6306802129714**
Status : **Answered**
Chosen Option : **1**

Q.40 Which of the following places receives rainfall from the Bay of Bengal branch of southwest monsoon?

- Ans** ☐ 1. Ratnagiri
☐ 2. Surat
☒ 3. Kolkata
☐ 4. Panaji

Question ID : **630680511048**
Option 1 ID : **6306801997323**
Option 2 ID : **6306801997325**
Option 3 ID : **6306801997322**
Option 4 ID : **6306801997324**
Status : **Answered**
Chosen Option : **3**

Q.41 The total number of bishops in a chess game is ____.

- Ans
- ☒ 1. one
 - ☐ 2. two
 - ☐ 3. three
 - ☒ 4. four

Question ID : 630680430544
 Option 1 ID : 6306801680197
 Option 2 ID : 6306801680198
 Option 3 ID : 6306801680199
 Option 4 ID : 6306801680200
 Status : Answered
 Chosen Option : 2

Q.42 In plant cells, which organelle is responsible for photosynthesis?

- Ans
- ☒ 1. Chloroplast
 - ☐ 2. Mitochondrion
 - ☐ 3. Lysosome
 - ☐ 4. Peroxisome

Question ID : 630680411646
 Option 1 ID : 6306801605419
 Option 2 ID : 6306801605420
 Option 3 ID : 6306801605421
 Option 4 ID : 6306801605422
 Status : Answered
 Chosen Option : 1

Q.43 Sange Meel Se Mulaqat is a documentary on the life of _____, directed by Gautam Ghosh.

- Ans
- ☒ 1. Pandit Bismillah Khan
 - ☐ 2. Pandit Bhimsen Joshi
 - ☐ 3. Pandit Biswajit Roy Chowdhury
 - ☐ 4. Pandit Ram Narayan

Question ID : 630680701402
 Option 1 ID : 6306802747548
 Option 2 ID : 6306802747545
 Option 3 ID : 6306802747547
 Option 4 ID : 6306802747546
 Status : Answered
 Chosen Option : 3

Q.44 Which organelle contains enzymes for digesting cellular waste and foreign materials?

- Ans
- ☐ 1. Nucleus
 - ☐ 2. Mitochondrion
 - ☒ 3. Lysosome
 - ☐ 4. Vacuole

Question ID : 630680411632
 Option 1 ID : 6306801605363
 Option 2 ID : 6306801605364
 Option 3 ID : 6306801605365
 Option 4 ID : 6306801605366
 Status : Answered
 Chosen Option : 3

Q.45 The National Defense Fund is governed by an Executive Committee, which is chaired by the:

- Ans** ☒ 1. President
☒ 2. Prime Minister
☒ 3. Vice President
☒ 4. Defence Minister

Question ID : **630680493649**
Option 1 ID : **6306801928324**
Option 2 ID : **6306801928325**
Option 3 ID : **6306801928326**
Option 4 ID : **6306801928327**
Status : **Answered**
Chosen Option : **1**

Q.46 You grew a small plant in a planter. On maturity, you observed that the plant has green stem, very few branches and the stem is so soft that you could easily break it with your hands. Under which category will you put this plant?

- Ans** ☒ 1. Tree
☒ 2. Creeper
☒ 3. Shrub
☒ 4. Herb

Question ID : **630680403058**
Option 1 ID : **6306801571801**
Option 2 ID : **6306801571802**
Option 3 ID : **6306801571803**
Option 4 ID : **6306801571804**
Status : **Answered**
Chosen Option : **4**

Q.47 Which compound is used to neutralise fatty acids and convert them into salts in a process called saponification?

- Ans** ☒ 1. Sodium acetate
☒ 2. Sodium chlorate
☒ 3. Sodium fluoride
☒ 4. Sodium hydroxide

Question ID : **630680403012**
Option 1 ID : **6306801571619**
Option 2 ID : **6306801571622**
Option 3 ID : **6306801571621**
Option 4 ID : **6306801571620**
Status : **Answered**
Chosen Option : **1**

Q.48 Which of the following Ministers introduced the Central Goods and Services Tax (Amendment) Bill 2023 and the Integrated Goods and Services Tax (Amendment) Bill 2023 in the Lok Sabha?

- Ans** ✓ 1. Nirmala Sitharaman
✗ 2. Harpal Singh
✗ 3. Bikram Keshari Arukha
✗ 4. Manohar Lal Khattar

Question ID : 630680469607
Option 1 ID : 6306801834191
Option 2 ID : 6306801834192
Option 3 ID : 6306801834193
Option 4 ID : 6306801834194
Status : Answered
Chosen Option : 1

Q.49 Which of the following is/are a gymnosperm?

- Ans** ✓ 1. Cycas
✗ 2. Mosses
✗ 3. Rose
✗ 4. Fern

Question ID : 630680542129
Option 1 ID : 6306802119457
Option 2 ID : 6306802119459
Option 3 ID : 6306802119458
Option 4 ID : 6306802119460
Status : Answered
Chosen Option : 1

Q.50 The particles that form the main part of the nucleus of an atom are together known as:

- Ans** ✗ 1. isotopes
✗ 2. ions
✗ 3. electrons
✓ 4. nucleons

Question ID : 630680401910
Option 1 ID : 6306801567388
Option 2 ID : 6306801567387
Option 3 ID : 6306801567386
Option 4 ID : 6306801567389
Status : Answered
Chosen Option : 4

Q.1 In the vapour compression refrigeration cycle with dry saturated vapour after compression, the entropy at the end of compression is the same as the _____.

- Ans
- ☒ 1. entropy at the end of expansion
 - ☒ 2. entropy at the end of evaporation
 - ☒ 3. entropy at the end of condensation
 - ☒ 4. entropy at the start of expansion

Question ID : 630680506398
Option 1 ID : 6306801979106
Option 2 ID : 6306801979107
Option 3 ID : 6306801979105
Option 4 ID : 6306801979108
Status : Answered
Chosen Option : 1

Q.2 If N is the speed of the compressor in RPM, then what will be the isentropic power for a double acting compressor? Where W = work required by the compressor.

- Ans
- ☒ 1. $WN/60$
 - ☒ 2. WN
 - ☒ 3. $2WN/60$
 - ☒ 4. $WN/2$

Question ID : 630680506433
Option 1 ID : 6306801979246
Option 2 ID : 6306801979245
Option 3 ID : 6306801979247
Option 4 ID : 6306801979248
Status : Answered
Chosen Option : 3

Q.3 In a Hartnell governor, if a spring of greater stiffness is used, then the governor will be:

- Ans
- ☒ 1. less sensitive
 - ☒ 2. insensitive
 - ☒ 3. Isochronous
 - ☒ 4. more sensitive

Question ID : 630680335145
Option 1 ID : 6306801303798
Option 2 ID : 6306801303800
Option 3 ID : 6306801303799
Option 4 ID : 6306801303797
Status : Answered
Chosen Option : 1

Q.4 Name the welding defect in which we observe a long and continuous visual separation line between the base metal and the heat affected zone.

- Ans**
- ☒ 1. Incomplete fusion
 - ☒ 2. Lamellar tearing
 - ☒ 3. Undercut
 - ☒ 4. Hot cracking

Question ID : 630680211307

Option 1 ID : 630680819377

Option 2 ID : 630680819375

Option 3 ID : 630680819378

Option 4 ID : 630680819376

Status : Answered

Chosen Option : 4

Q.5 Which of the following statements is true about hit and miss governing?

- Ans**
- ☒ 1. In this method of governing, none of the cycles should be idle throughout the running of engine.
 - ☒ 2. It is a suitable governing method for heavy IC engines.
 - ☒ 3. It increases the efficiency of engine.
 - ☒ 4. This method of governing requires higher weight of flywheel.

Question ID : 630680502674

Option 1 ID : 6306801964281

Option 2 ID : 6306801964278

Option 3 ID : 6306801964279

Option 4 ID : 6306801964280

Status : Answered

Chosen Option : 1

Q.6 Which of the following is used to increase the temperature of steam above its saturation temperature?

- Ans**
- ☒ 1. Air pre-heater
 - ☒ 2. Economiser
 - ☒ 3. Steam dryer
 - ☒ 4. Superheater

Question ID : 630680502707

Option 1 ID : 6306801964411

Option 2 ID : 6306801964410

Option 3 ID : 6306801964413

Option 4 ID : 6306801964412

Status : Answered

Chosen Option : 4

Q.7 Under which of the following conditions does the Francis turbine operate?

- Ans**
- ☒ 1. Medium head and medium discharge
 - ☒ 2. High head and high discharge
 - ☒ 3. High head and low discharge
 - ☒ 4. Low head and high discharge

Question ID : 630680193066

Option 1 ID : 630680748802

Option 2 ID : 630680748804

Option 3 ID : 630680748801

Option 4 ID : 630680748803

Status : Answered

Chosen Option : 1

Q.8 A drill bit of diameter 25 mm has cutting speed of 15.7 m/min. What is the speed of rotation of drill?

- Ans
- ☒ 1. 200 RPM
 - ☐ 2. 314 RPM
 - ☐ 3. 500 RPM
 - ☐ 4. 400 RPM

Question ID : 630680503251
Option 1 ID : 6306801966590
Option 2 ID : 6306801966592
Option 3 ID : 6306801966591
Option 4 ID : 6306801966589
Status : Answered
Chosen Option : 1

Q.9 In the process of brazing, the filler metal is drawn into the joint by means of _____.

- Ans
- ☐ 1. Low viscosity
 - ☐ 2. Surface tension
 - ☒ 3. Capillary action
 - ☐ 4. High diffusion

Question ID : 630680205053
Option 1 ID : 630680795077
Option 2 ID : 630680795075
Option 3 ID : 630680795076
Option 4 ID : 630680795078
Status : Answered
Chosen Option : 3

Q.10 Which of the following is NOT the application for steam in industry?

- Ans
- ☐ 1. Heating
 - ☐ 2. Propulsion
 - ☐ 3. Humidification
 - ☒ 4. De-atomisation

Question ID : 630680348139
Option 1 ID : 6306801354845
Option 2 ID : 6306801354846
Option 3 ID : 6306801354847
Option 4 ID : 6306801354848
Status : Answered
Chosen Option : 4

Q.11 Which of the following impellers are best suited to medium sized pumps with a small amount of soft solids?

- Ans
- ☐ 1. Open impellers
 - ☐ 2. Closed impellers
 - ☒ 3. Semi-open impellers
 - ☐ 4. Semi-closed impellers

Question ID : 630680508726
Option 1 ID : 6306801988204
Option 2 ID : 6306801988205
Option 3 ID : 6306801988207
Option 4 ID : 6306801988206
Status : Answered
Chosen Option : 3

Q.12 In the porter governor, the main constituent of controlling force is:

- Ans
- ☐ 1. friction force
 - ☐ 2. spring force
 - ☐ 3. mass of flyball
 - ☒ 4. mass of sleeve

Question ID : 630680509132
Option 1 ID : 6306801989824
Option 2 ID : 6306801989823
Option 3 ID : 6306801989825
Option 4 ID : 6306801989826
Status : Answered
Chosen Option : 3

Q.13 If the power output of a hydraulic turbine is 100 kW and overall efficiency is 50%, then what will be the power supplied at the inlet of the turbine?

- Ans
- ☐ 1. 150 kW
 - ☐ 2. 250 kW
 - ☐ 3. 50 kW
 - ☒ 4. 200 kW

Question ID : 630680508632
Option 1 ID : 6306801987829
Option 2 ID : 6306801987831
Option 3 ID : 6306801987828
Option 4 ID : 6306801987830
Status : Answered
Chosen Option : 4

Q.14 The casting process that employs a permanent metal or ceramic mould, is _____.

- Ans
- ☐ 1. sand casting
 - ☒ 2. die casting
 - ☐ 3. investment casting
 - ☐ 4. centrifugal casting

Question ID : 630680503205
Option 1 ID : 6306801966405
Option 2 ID : 6306801966407
Option 3 ID : 6306801966406
Option 4 ID : 6306801966408
Status : Answered
Chosen Option : 2

Q.15 A cantilever beam of length 2 m is subjected to a point load of 3 kN at a distance of 2 m and is subjected to a UDL of 3 kN/m for a total distance of 1.5 m from the fixed end. Calculate the shear force and bending moment at the fixed end.

- Ans
- ☐ 1. -5.75 kN, -5.545 kN-m
 - ☐ 2. -10 kN, 15.254 kN-m
 - ☐ 3. -12.5 kN, -8.547 kN-m
 - ☒ 4. -7.5 kN, -9.375 kN-m

Question ID : 630680205958
Option 1 ID : 630680798637
Option 2 ID : 630680798636
Option 3 ID : 630680798638
Option 4 ID : 630680798635
Status : Answered
Chosen Option : 1

Q.16 In actual air-conditioning applications, the heat rejection factor depends upon the:

- Ans ☒ 1. evaporator and condenser temperatures
- ☐ 2. condenser temperature
- ☐ 3. rate of flow
- ☐ 4. evaporator temperature

Question ID : 630680506427
 Option 1 ID : 6306801979224
 Option 2 ID : 6306801979221
 Option 3 ID : 6306801979223
 Option 4 ID : 6306801979222
 Status : Answered
 Chosen Option : 2

Q.17 The conventional depth of a cut considered in the parting operation using a lathe is _____.

- Ans ☐ 1. 20 mm to 25 mm
- ☐ 2. 1 mm to 2 mm
- ☒ 3. 3 mm to 10 mm
- ☐ 4. 12 mm to 16 mm

Question ID : 630680205078
 Option 1 ID : 630680795178
 Option 2 ID : 630680795176
 Option 3 ID : 630680795175
 Option 4 ID : 630680795177
 Status : Answered
 Chosen Option : 2

Q.18 If h_1 and h_2 are enthalpies at the inlet and the outlet, respectively and C_1 and C_2 are inlet and outlet velocities, respectively, then which of the following statements is INCORRECT about the equation $(\frac{C_2^2 - C_1^2}{2} = h_1 - h_2)$ for steam nozzles?

- Ans ☒ 1. The equation considers frictional losses in the steam nozzle.
- ☐ 2. The equation is the steady flow energy equation for steam nozzles.
- ☐ 3. The flow is adiabatic in steam nozzles.
- ☐ 4. There is no mechanical work done by the steam nozzle.

Question ID : 630680502641
 Option 1 ID : 6306801964149
 Option 2 ID : 6306801964146
 Option 3 ID : 6306801964147
 Option 4 ID : 6306801964148
 Status : Answered
 Chosen Option : 1

Q.19 Which of the following statements is **INCORRECT** about the air pre-heater in a steam boiler?

- Ans** ☒ 1. In the regenerative type of air pre-heater, heat from flue gases is transferred to air through an intermediate heat storage medium.
- ☒ 2. In the tubular type of air pre-heater, hot flue gases flow through the outside of the tubes of the air pre-heater.
- ☒ 3. In the tubular type of air pre-heater, hot flue gases flow in the direction opposite to that of air travel.
- ☒ 4. Use of the air pre-heater leads to less atmospheric pollution.

Question ID : 630680502656
 Option 1 ID : 6306801964207
 Option 2 ID : 6306801964206
 Option 3 ID : 6306801964209
 Option 4 ID : 6306801964208
 Status : **Answered**
 Chosen Option : 4

Q.20 In case of a four-stroke diesel engine, the valve timing diagram is expressed in terms of _____.

- Ans** ☒ 1. the time taken for opening and closing of the valve with respect to TDC and BDC
- ☒ 2. the time taken for opening and closing of the valve with respect to only TDC
- ☒ 3. the degree of the crank angle at the time of opening and closing of the valve
- ☒ 4. the linear movement of the piston with respect to TDC and BDC

Question ID : 630680502711
 Option 1 ID : 6306801964426
 Option 2 ID : 6306801964427
 Option 3 ID : 6306801964428
 Option 4 ID : 6306801964429
 Status : **Answered**
 Chosen Option : 3

Q.21 Enthalpy of a system in an open flow is given by $h = u + pv$, where pv is the _____.

- Ans** ☒ 1. momentum energy
- ☒ 2. moving boundary work
- ☒ 3. flow work
- ☒ 4. external work

Question ID : 630680503243
 Option 1 ID : 6306801966559
 Option 2 ID : 6306801966557
 Option 3 ID : 6306801966558
 Option 4 ID : 6306801966560
 Status : **Answered**
 Chosen Option : 3

Q.22 The heat transfer during constant pressure heating of a gas in a cylinder containing a sliding piston is equal to _____.

- Ans
- ☐ 1. the change in internal energy
 - ☐ 2. the work done by the piston
 - ☐ 3. zero
 - ☒ 4. the change in enthalpy

Question ID : 630680503275
Option 1 ID : 6306801966685
Option 2 ID : 6306801966687
Option 3 ID : 6306801966688
Option 4 ID : 6306801966686
Status : Answered
Chosen Option : 4

Q.23 The Kelvin-Planck statement of the Second Law states that _____.

- Ans
- ☐ 1. heat cannot flow spontaneously from a colder body to a hotter body
 - ☐ 2. it is possible to convert all the heat absorbed from a single reservoir into work
 - ☒ 3. it is impossible to convert all the heat absorbed from a single reservoir into work
 - ☐ 4. work can be converted completely into heat

Question ID : 630680503229
Option 1 ID : 6306801966503
Option 2 ID : 6306801966502
Option 3 ID : 6306801966501
Option 4 ID : 6306801966504
Status : Answered
Chosen Option : 3

Q.24 Which of the following statements is INCORRECT about no slip boundary condition?

- Ans
- ☐ 1. The no slip condition can be defined for viscous flows.
 - ☐ 2. During no slip condition the fluid velocity at all fluid–solid boundaries is equal to that of the solid boundary.
 - ☒ 3. In case of the no slip boundary condition at a fixed solid boundary, the fluid will have some velocity relative to the boundary.
 - ☐ 4. In case of the no slip boundary condition at a fixed solid boundary, the fluid will have zero velocity.

Question ID : 630680199006
Option 1 ID : 630680771754
Option 2 ID : 630680771751
Option 3 ID : 630680771752
Option 4 ID : 630680771753
Status : Answered
Chosen Option : 3

Q.25 The casting process that employs a wax pattern coated with a ceramic shell to make the mould, is _____.

- Ans
- ☐ 1. centrifugal casting
 - ☐ 2. die casting
 - ☒ 3. investment casting
 - ☐ 4. sand casting

Question ID : 630680503204
Option 1 ID : 6306801966404
Option 2 ID : 6306801966403
Option 3 ID : 6306801966402
Option 4 ID : 6306801966401
Status : Answered
Chosen Option : 3

Q.26 The mist lubrication system is generally used for _____.

- Ans
- ☐ 1. four stroke cycle engine
 - ☐ 2. air refrigeration cycle engine
 - ☒ 3. two stroke cycle engine
 - ☐ 4. gas engine

Question ID : 630680502667
Option 1 ID : 6306801964250
Option 2 ID : 6306801964252
Option 3 ID : 6306801964251
Option 4 ID : 6306801964253
Status : Answered
Chosen Option : 3

Q.27 The volumetric efficiency of a single-stage reciprocating air compressor is the ratio of the:

- Ans
- ☐ 1. displacement of the compressor to the free air delivered
 - ☐ 2. swept volume to the effective swept volume
 - ☐ 3. swept volume to the clearance volume
 - ☒ 4. free air delivered to the displacement of the compressor

Question ID : 630680181651
Option 1 ID : 630680703847
Option 2 ID : 630680703849
Option 3 ID : 630680703848
Option 4 ID : 630680703846
Status : Answered
Chosen Option : 1

Q.28 The brake power of an engine can be measured by using a rope brake dynamometer. It is given by the expression: (Where, 'D' is diameter of the brake drum; 'W' is the weight and 'S' is the spring scale reading, 'N' Speed in rpm)

- Ans
- ☒ 1. $\pi DN (W + S)$
 - ☒ 2. $DN (W + S)$
 - ☒ 3. $DN (W - S)$
 - ☒ 4. $\pi DN (W - S)$

Question ID : 630680386536
 Option 1 ID : 6306801506854
 Option 2 ID : 6306801506857
 Option 3 ID : 6306801506856
 Option 4 ID : 6306801506855
 Status : Answered
 Chosen Option : 1

Q.29 When the compressor and motor operate on the same shaft and are enclosed in a common casing, they are known as _____.

- Ans
- ☒ 1. reciprocating compressors
 - ☒ 2. axial compressors
 - ☒ 3. hermetic sealed compressors
 - ☒ 4. centrifugal compressors

Question ID : 630680506440
 Option 1 ID : 6306801979273
 Option 2 ID : 6306801979276
 Option 3 ID : 6306801979275
 Option 4 ID : 6306801979274
 Status : Answered
 Chosen Option : 3

Q.30 A cyclic heat engine takes 40 kJ of heat from a 100°C temperature reservoir. If it gives 40 kJ of work, then which of the following statements is correct?

- Ans
- ☒ 1. The engine violates the Kelvin Planck's statement
 - ☒ 2. The engine violates 1st law of thermodynamics as well as Kelvin Planck's statement.
 - ☒ 3. The engine violates the Clausius statement.
 - ☒ 4. The engine violates the 1st law of thermodynamics.

Question ID : 630680503257
 Option 1 ID : 6306801966614
 Option 2 ID : 6306801966616
 Option 3 ID : 6306801966615
 Option 4 ID : 6306801966613
 Status : Answered
 Chosen Option : 1

Q.31 If there is flow of real fluid at fixed boundary, then no-slip condition:

- Ans
- ☒ 1. depends on the type of flow
 - ☒ 2. may or may not occur
 - ☒ 3. occurs
 - ☒ 4. does not occur

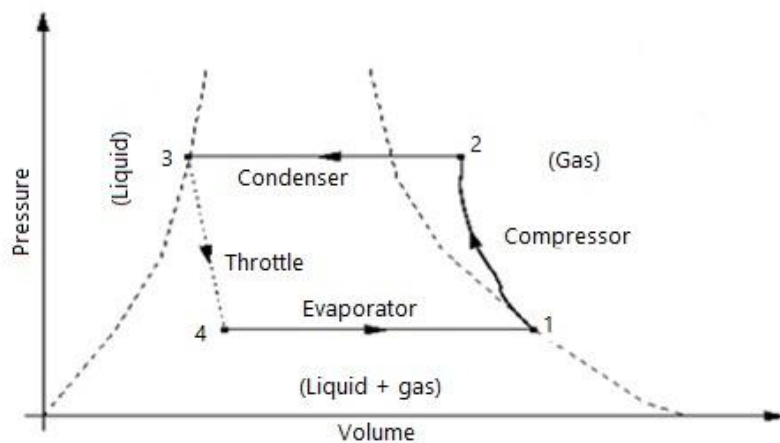
Question ID : 630680508622
 Option 1 ID : 6306801987791
 Option 2 ID : 6306801987790
 Option 3 ID : 6306801987789
 Option 4 ID : 6306801987788
 Status : Answered
 Chosen Option : 3

Q.32 Total gradient line (or total energy line) is parallel to hydraulic gradient line with a vertical distance of _____. (The symbols have their usual meanings.)

- Ans
- ☒ 1. $\frac{v^2}{2g}$
 - ☒ 2. $\frac{p}{w}$
 - ☒ 3. $\left(\frac{p}{w}\right) + \left(\frac{v^2}{2g}\right) + z$
 - ☒ 4. $\left(\frac{p}{w}\right) + z$

Question ID : 630680198989
 Option 1 ID : 630680771685
 Option 2 ID : 630680771683
 Option 3 ID : 630680771686
 Option 4 ID : 630680771684
 Status : Answered
 Chosen Option : 1

Q.33 The given figure shows the p-V diagram of:



- Ans
- ☒ 1. vapour absorption refrigeration system
 - ☒ 2. aircraft refrigeration system
 - ☒ 3. vapour compression refrigeration system
 - ☒ 4. Electrolux refrigeration system

Question ID : 630680332164
 Option 1 ID : 6306801292102
 Option 2 ID : 6306801292104
 Option 3 ID : 6306801292101
 Option 4 ID : 6306801292103
 Status : Answered
 Chosen Option : 3

Q.34 The heat rejection in Stirling cycle takes place at:

- Ans
- ☒ 1. constant temperature
 - ☒ 2. constant pressure
 - ☒ 3. constant volume
 - ☒ 4. constant enthalpy

Question ID : 630680412680
 Option 1 ID : 6306801609545
 Option 2 ID : 6306801609544
 Option 3 ID : 6306801609543
 Option 4 ID : 6306801609546
 Status : Answered
 Chosen Option : 3

Q.35 Which of the following statements is true about an SI engine in comparison to a CI engine?

- Ans
- ☒ 1. The air-fuel ratio is high.
 - ☒ 2. The cost of running is comparatively low.
 - ☒ 3. The operating speed is very high.
 - ☒ 4. The compression ratio is comparatively high.

Question ID : 630680502709
 Option 1 ID : 6306801964418
 Option 2 ID : 6306801964421
 Option 3 ID : 6306801964420
 Option 4 ID : 6306801964419
 Status : Answered
 Chosen Option : 3

Q.36 Which part of the arc welding element/equipment is consumable?

- Ans ☒ 1. Electrode holder
☒ 2. AC/DC transformer
☒ 3. Electrical cables
☒ 4. Electrode

Question ID : 630680205050

Option 1 ID : 630680795065

Option 2 ID : 630680795064

Option 3 ID : 630680795063

Option 4 ID : 630680795066

Status : Answered

Chosen Option : 4

Q.37 Variation of power input with speed at constant discharge in case of a centrifugal pump is _____.

- Ans ☒ 1. linear
☒ 2. logarithmic
☒ 3. cubic
☒ 4. parabolic

Question ID : 630680198968

Option 1 ID : 630680771599

Option 2 ID : 630680771602

Option 3 ID : 630680771601

Option 4 ID : 630680771600

Status : Answered

Chosen Option : 3

Q.38 When the fluid pressure is measured above the absolute zero pressure, the measured pressure is known as _____.

- Ans ☒ 1. vacuum pressure
☒ 2. absolute pressure
☒ 3. atmospheric pressure
☒ 4. gauge pressure

Question ID : 630680193056

Option 1 ID : 630680748764

Option 2 ID : 630680748762

Option 3 ID : 630680748761

Option 4 ID : 630680748763

Status : Answered

Chosen Option : 4

Q.39 Carbon steel contains a carbon percentage of about 0.21%. The possible type of carbon steel is _____.

- Ans ☒ 1. stainless steel
☒ 2. medium carbon steel
☒ 3. mild steel
☒ 4. alloy steel

Question ID : 630680191553

Option 1 ID : 630680742836

Option 2 ID : 630680742834

Option 3 ID : 630680742835

Option 4 ID : 630680742833

Status : Answered

Chosen Option : 3

Q.40 In the battery ignition system for SI engines, with increase in the speed, the _____.

- Ans
- ☐ 1. sparking voltage remains the same
 - ☒ 2. sparking voltage drops
 - ☐ 3. sparking voltage increases
 - ☐ 4. sparking voltage first increases and then becomes constant

Question ID : 630680502730
Option 1 ID : 6306801964504
Option 2 ID : 6306801964503
Option 3 ID : 6306801964502
Option 4 ID : 6306801964505
Status : Answered
Chosen Option : 3

Q.41 What will be the value of the axial thrust on a wheel when there is no blade friction?

- Ans
- ☐ 1. Maximum
 - ☐ 2. Minimum
 - ☒ 3. Zero
 - ☐ 4. Equal to the tangential force on the blade

Question ID : 630680502690
Option 1 ID : 6306801964342
Option 2 ID : 6306801964343
Option 3 ID : 6306801964344
Option 4 ID : 6306801964345
Status : Answered
Chosen Option : 4

Q.42 The location of centre of gravity for a sphere of radius (R) from the geometrical centre is at a distance of:

- Ans
- ☒ 1. 0
 - ☐ 2. 0.5R
 - ☐ 3. 0.25R
 - ☐ 4. R

Question ID : 630680509611
Option 1 ID : 6306801991668
Option 2 ID : 6306801991666
Option 3 ID : 6306801991667
Option 4 ID : 6306801991665
Status : Answered
Chosen Option : 3

Q.43 When pressure is measured above the atmospheric pressure, it is termed as:

- Ans
- ☐ 1. stagnation pressure
 - ☐ 2. absolute pressure
 - ☒ 3. gauge pressure
 - ☐ 4. vacuum pressure

Question ID : 630680508636
Option 1 ID : 6306801987847
Option 2 ID : 6306801987844
Option 3 ID : 6306801987846
Option 4 ID : 6306801987845
Status : Answered
Chosen Option : 3

Q.44 What is the main benefit of double volute casing over single volute casing of a centrifugal pump?

- Ans
- ☒ 1. Less power requirement
 - ☒ 2. Radial load balancing
 - ☐ 3. High efficiency
 - ☐ 4. High discharge

Question ID : 630680198970

Option 1 ID : 630680771608

Option 2 ID : 630680771607

Option 3 ID : 630680771610

Option 4 ID : 630680771609

Status : Answered

Chosen Option : 4

Q.45 In which of the following compressors, the top cover is bolted to the compressor housing instead of the welded steel shell?

- Ans
- ☐ 1. Axial compressor
 - ☐ 2. Open compressor
 - ☐ 3. Hermetic sealed compressor
 - ☒ 4. Semi-hermetic sealed compressor

Question ID : 630680506436

Option 1 ID : 6306801979260

Option 2 ID : 6306801979259

Option 3 ID : 6306801979257

Option 4 ID : 6306801979258

Status : Answered

Chosen Option : 4

Q.46 The distance of the centre of pressure from the free surface of the liquid is independent of the:

- Ans
- ☐ 1. depth of the centre of gravity from the free surface
 - ☒ 2. density of liquid
 - ☐ 3. area exposed to the liquid
 - ☐ 4. moment of inertia about the centre of gravity

Question ID : 630680508740

Option 1 ID : 6306801988261

Option 2 ID : 6306801988262

Option 3 ID : 6306801988260

Option 4 ID : 6306801988263

Status : Answered

Chosen Option : 2

Q.47 Which of the following statements is true about quantity governing?

- Ans
- ☒ 1. The mixture strength remains the same with variation in the speed of the engine.
 - ☐ 2. The compression ratio varies with charge supplied.
 - ☐ 3. Due to quantity governing, the engine efficiency is altered.
 - ☐ 4. This method is mostly preferred for diesel engines.

Question ID : 630680502681

Option 1 ID : 6306801964306

Option 2 ID : 6306801964308

Option 3 ID : 6306801964309

Option 4 ID : 6306801964307

Status : Answered

Chosen Option : 1

Q.48 A complex geometry is placed in the X-Y plane. If the geometry is found to be symmetrical about the Y-axis, then which of the following is correct to calculate the centre of gravity?

- Ans**
- ☒ 1. X co-ordinate is zero
 - ☐ 2. X co-ordinate is a non-zero value
 - ☐ 3. Z co-ordinate is a non-zero value
 - ☐ 4. Y co-ordinate is zero

Question ID : 630680509603
Option 1 ID : 6306801991635
Option 2 ID : 6306801991634
Option 3 ID : 6306801991636
Option 4 ID : 6306801991633
Status : **Answered**
Chosen Option : 4

Q.49 Which of the following is correct for Pelton wheel?
Where, α & β = guide blade angle at inlet and outlet, respectively
 Θ & ϕ = vane angle at inlet and outlet, respectively

- Ans**
- ☒ 1. $\alpha = 0^\circ$ and $\Theta = 0^\circ$
 - ☐ 2. $\beta = 0^\circ$ and $\phi = 0^\circ$
 - ☐ 3. $\alpha = 0^\circ$ and $\beta = 0^\circ$
 - ☐ 4. $\alpha = 0^\circ$ and $\phi = 0^\circ$

Question ID : 630680508732
Option 1 ID : 6306801988231
Option 2 ID : 6306801988229
Option 3 ID : 6306801988230
Option 4 ID : 6306801988228
Status : **Answered**
Chosen Option : 1

Q.50 'No slip condition' at a fixed surface is applicable to the flow of which type of fluids?

- Ans**
- ☒ 1. All real fluids
 - ☐ 2. Only Newtonian fluids
 - ☐ 3. All non-Newtonian fluids
 - ☐ 4. Only ideal fluids

Question ID : 630680508717
Option 1 ID : 6306801988170
Option 2 ID : 6306801988168
Option 3 ID : 6306801988169
Option 4 ID : 6306801988171
Status : **Answered**
Chosen Option : 4

Q.51 Which of the following is NOT a lathe turning operation?

- Ans ☒ 1. Sawing
- ☐ 2. Grooving
- ☐ 3. Facing
- ☐ 4. Threading

Question ID : 630680503220
 Option 1 ID : 6306801966467
 Option 2 ID : 6306801966468
 Option 3 ID : 6306801966465
 Option 4 ID : 6306801966466
 Status : Answered
 Chosen Option : 1

Q.52 Which of the following is the assumption of Air Standard Cycles?

- Ans ☐ 1. The working medium has variable specific heat with varying temperature
- ☒ 2. Some heat is assumed to be rejected to a constant low temperature sink
- ☐ 3. The working medium is assumed to be a real gas
- ☐ 4. Heat is assumed to be supplied from chemical reactions during the cycle

Question ID : 630680502664
 Option 1 ID : 6306801964240
 Option 2 ID : 6306801964241
 Option 3 ID : 6306801964238
 Option 4 ID : 6306801964239
 Status : Answered
 Chosen Option : 4

Q.53 For the laminar flow through circular pipes, shear stress distribution across a section is _____.

- Ans ☐ 1. logarithmic
- ☒ 2. linear
- ☐ 3. cubic
- ☐ 4. parabolic

Question ID : 630680193038
 Option 1 ID : 630680748691
 Option 2 ID : 630680748689
 Option 3 ID : 630680748692
 Option 4 ID : 630680748690
 Status : Answered
 Chosen Option : 2

Q.54 Which of the following is NOT a part of a bucket type steam trap?

- Ans ☐ 1. Guide tube
- ☐ 2. Float
- ☐ 3. Spiral vane
- ☒ 4. Injector

Question ID : 630680502654
 Option 1 ID : 6306801964199
 Option 2 ID : 6306801964198
 Option 3 ID : 6306801964200
 Option 4 ID : 6306801964201
 Status : Answered
 Chosen Option : 4

Q.55 Which of the following evaporators are used for refrigeration units of 2 to 250 TR capacity?

- Ans**
- ☐ 1. Flooded shell and tube evaporators
 - ☐ 2. Shell and coil evaporators
 - ☐ 3. Plate evaporators
 - ☒ 4. Dry expansion shell and tube evaporators

Question ID : **630680506419**
Option 1 ID : **6306801979189**
Option 2 ID : **6306801979190**
Option 3 ID : **6306801979191**
Option 4 ID : **6306801979192**
Status : **Answered**
Chosen Option : **1**

Q.56 Which of the following statements is correct about the location of centre of pressure?

- Ans**
- ☐ 1. Centre of pressure may lie at any location irrespective of the centre of gravity of the vertically immersed surface.
 - ☐ 2. Centre of pressure lies above the centre of gravity of the vertically immersed surface.
 - ☐ 3. Centre of pressure coincides with the centre of gravity of the vertically immersed surface.
 - ☒ 4. Centre of pressure lies below the centre of gravity of the vertically immersed surface.

Question ID : **630680199002**
Option 1 ID : **630680771738**
Option 2 ID : **630680771736**
Option 3 ID : **630680771737**
Option 4 ID : **630680771735**
Status : **Answered**
Chosen Option : **4**

Q.57 Which of the following gas combination is commonly used in oxyfuel-gas welding?

- Ans**
- ☐ 1. Oxygen and Nitrogen
 - ☐ 2. Oxygen and CO₂
 - ☐ 3. Oxygen and H₂
 - ☒ 4. Oxygen and acetylene

Question ID : **630680503272**
Option 1 ID : **6306801966676**
Option 2 ID : **6306801966674**
Option 3 ID : **6306801966675**
Option 4 ID : **6306801966673**
Status : **Answered**
Chosen Option : **4**

Q.58 In a milling machine, bulky workpieces of irregular shapes are clamped directly on the milling machine table by using _____.

- Ans
- ☐ 1. V-blocks
 - ☐ 2. angle plates
 - ☐ 3. swivel vise
 - ☒ 4. T-bolts and clamps

Question ID : 630680205087

Option 1 ID : 630680795213

Option 2 ID : 630680795211

Option 3 ID : 630680795214

Option 4 ID : 630680795212

Status : Answered

Chosen Option : 4

Q.59 The mass transfer process in an open system is commonly modelled using the _____ equation.

- Ans
- ☐ 1. zeroth law of thermodynamics
 - ☐ 2. third law of thermodynamics
 - ☐ 3. second law of thermodynamics
 - ☒ 4. mass balance

Question ID : 630680503178

Option 1 ID : 6306801966297

Option 2 ID : 6306801966299

Option 3 ID : 6306801966298

Option 4 ID : 6306801966300

Status : Answered

Chosen Option : 4

Q.60 Which of the following is NOT a frequent pattern-making process in manufacturing?

- Ans
- ☐ 1. Match plate pattern
 - ☒ 2. Welding pattern
 - ☐ 3. Investment pattern
 - ☐ 4. Sweep pattern

Question ID : 630680503200

Option 1 ID : 6306801966386

Option 2 ID : 6306801966388

Option 3 ID : 6306801966387

Option 4 ID : 6306801966385

Status : Answered

Chosen Option : 2

Q.61 What will be the power lost in friction assuming uniform pressure theory, when a vertical shaft of 100 mm diameter rotating at 150 r.p.m. rests on a flat end foot step bearing? The coefficient of friction is equal to 0.05 and the shaft carries a vertical load of 15 KN.

- Ans** ✓ 1. 392.7 W
✗ 2. 392.7 KW
✗ 3. 39.27 KW
✗ 4. 39.27 W

Question ID : 630680227834
Option 1 ID : 630680884013
Option 2 ID : 630680884011
Option 3 ID : 630680884012
Option 4 ID : 630680884014
Status : **Not Answered**
Chosen Option : --

Q.62 Determine the stress induced in the steel bars of reinforced concrete structure, if the modular ratio for steel and concrete is 14.5 and the stress applied on the concrete is 2.5 MPa.

- Ans** ✗ 1. 24.50 MPa
✗ 2. 42.75 MPa
✗ 3. 30.25 MPa
✓ 4. 36.25 MPa

Question ID : 630680327982
Option 1 ID : 6306801275898
Option 2 ID : 6306801275899
Option 3 ID : 6306801275900
Option 4 ID : 6306801275897
Status : **Answered**
Chosen Option : 2

Q.63 Which of the following prevents the high-pressure vapour refrigerant from flowing back to the evaporator in rotating blade type rotary compressor?

- Ans** ✗ 1. Pressure valve
✗ 2. Throttling valve
✓ 3. Check valve
✗ 4. Safety valve

Question ID : 630680506437
Option 1 ID : 6306801979264
Option 2 ID : 6306801979262
Option 3 ID : 6306801979263
Option 4 ID : 6306801979261
Status : **Answered**
Chosen Option : 3

Q.64 The manometric head of a centrifugal pump is independent of which of the following factors?

- Ans ☒ 1. Loss of head in impeller and casing
☒ 2. Frictional losses in bearings
☒ 3. Frictional head loss in suction pipe
☒ 4. Frictional head loss in delivery pipe

Question ID : 630680198966

Option 1 ID : 630680771591

Option 2 ID : 630680771593

Option 3 ID : 630680771592

Option 4 ID : 630680771594

Status : Answered

Chosen Option : 2

Q.65 A cooling system is essentially required in internal combustion engines to

- Ans ☒ 1. keep the engine very hot or very cool
☒ 2. remove heat at a faster rate when engine is hot
☒ 3. remove heat at a faster rate when the engine is cool
☒ 4. remove about 70% of heat generated in the combustion chamber

Question ID : 630680382380

Option 1 ID : 6306801490436

Option 2 ID : 6306801490435

Option 3 ID : 6306801490437

Option 4 ID : 6306801490438

Status : Answered

Chosen Option : 2

Q.66 Which of the following refrigeration lubricants were the first synthetic oils to be used in the refrigeration industry?

- Ans ☒ 1. Alkylbenzenes oil
☒ 2. Poly alkylene glycol oil
☒ 3. Polyol ester oil
☒ 4. Mineral oil

Question ID : 630680506418

Option 1 ID : 6306801979186

Option 2 ID : 6306801979188

Option 3 ID : 6306801979187

Option 4 ID : 6306801979185

Status : Answered

Chosen Option : 2

Q.67 The zeroth law of thermodynamics is based on the concept of _____.

- Ans ☒ 1. heat capacity
☒ 2. enthalpy
☒ 3. temperature
☒ 4. entropy

Question ID : 630680503210

Option 1 ID : 6306801966425

Option 2 ID : 6306801966428

Option 3 ID : 6306801966427

Option 4 ID : 6306801966426

Status : Answered

Chosen Option : 3

Q.68 Which of the following options is correct about four-stroke engines when compared with two-stroke engines?

Ans ☒ 1. Thermal efficiency is higher.

☐ 2. The thermodynamic cycle is completed in two strokes of the piston or in one revolution of the crankshaft.

☐ 3. Light weight and simplicity due to absence of the valve actuating mechanism, with the initial cost of the engine being low

☐ 4. Lower volumetric efficiency due to lesser time taken for mixture intake

Question ID : 630680181695

Option 1 ID : 630680704024

Option 2 ID : 630680704022

Option 3 ID : 630680704025

Option 4 ID : 630680704023

Status : Answered

Chosen Option : 1

Q.69 For the H-S diagram of a vapour compression refrigeration cycle, the specific enthalpy of the refrigerant coming out from the compressor _____.

Ans ☐ 1. may increase or decrease

☒ 2. increases

☐ 3. decreases

☐ 4. remains constant

Question ID : 630680506443

Option 1 ID : 6306801979288

Option 2 ID : 6306801979285

Option 3 ID : 6306801979286

Option 4 ID : 6306801979287

Status : Answered

Chosen Option : 4

Q.70 Which of the following is NOT a type of sawing?

Ans ☐ 1. Hack sawing

☒ 2. Straight sawing

☐ 3. Circular sawing

☐ 4. Band sawing

Question ID : 630680503248

Option 1 ID : 6306801966577

Option 2 ID : 6306801966579

Option 3 ID : 6306801966580

Option 4 ID : 6306801966578

Status : Answered

Chosen Option : 4

Q.71 What will not happen if the liquid has greater adhesion than cohesion?

- Ans
- ☐ 1. Liquid will tend to rise at the point of contact.
 - ☐ 2. Liquid will wet a solid surface.
 - ☐ 3. The angle of contact will be less than 90° in a small diameter tube.
 - ☒ 4. Liquid surface will be concave downwards in a small diameter tube.

Question ID : 630680198993

Option 1 ID : 630680771700

Option 2 ID : 630680771699

Option 3 ID : 630680771702

Option 4 ID : 630680771701

Status : Answered

Chosen Option : 4

Q.72 Which of the following machines violates the Kelvin Planck's statement?

- Ans
- ☐ 1. Reversible heat engine
 - ☐ 2. Irreversible heat engine
 - ☐ 3. PMM1
 - ☒ 4. PMM2

Question ID : 630680503256

Option 1 ID : 6306801966609

Option 2 ID : 6306801966610

Option 3 ID : 6306801966611

Option 4 ID : 6306801966612

Status : Answered

Chosen Option : 3

Q.73 The overall efficiency of the turbine is given by:

- Ans
- ☐ 1. overall efficiency = mechanical efficiency / hydraulic efficiency
 - ☐ 2. overall efficiency = mechanical efficiency² / hydraulic efficiency²
 - ☒ 3. overall efficiency = hydraulic efficiency $\times$ mechanical efficiency
 - ☐ 4. overall efficiency = hydraulic efficiency / mechanical efficiency

Question ID : 630680508612

Option 1 ID : 6306801987749

Option 2 ID : 6306801987750

Option 3 ID : 6306801987751

Option 4 ID : 6306801987748

Status : Answered

Chosen Option : 3

Q.74 For measurement of dryness fraction of steam, use of a separating calorimeter is suitable _____.

- Ans
- ☐ 1. when dryness fraction is greater than 1
 - ☐ 2. for any condition of steam
 - ☐ 3. when dryness fraction is greater than 0.95
 - ☒ 4. when dryness fraction is less than 0.95

Question ID : 630680502761

Option 1 ID : 6306801964629

Option 2 ID : 6306801964628

Option 3 ID : 6306801964626

Option 4 ID : 6306801964627

Status : Answered

Chosen Option : 3

Q.75 Isochronism in a governor is desirable when ____.

- Ans ☒ 1. the engine operates at low speeds
☒ 2. one speed is desired under one load
☒ 3. the engine is just getting started
☒ 4. the engine operates at high speeds

Question ID : 630680502673
Option 1 ID : 6306801964274
Option 2 ID : 6306801964277
Option 3 ID : 6306801964276
Option 4 ID : 6306801964275
Status : Answered
Chosen Option : 2

Q.76 Which of the following assertions related to drilling is true?

- Ans ☒ 1. Drilling is a process that uses a saw to make a hole in a workpiece.
☒ 2. Drilling is a process that uses a non-rotating tool to make a hole in a workpiece.
☒ 3. Drilling is a process that uses a rotating tool to make a hole in a workpiece.
☒ 4. Drilling is a process that uses a hammer to make a hole in a workpiece.

Question ID : 630680503223
Option 1 ID : 6306801966479
Option 2 ID : 6306801966478
Option 3 ID : 6306801966477
Option 4 ID : 6306801966480
Status : Answered
Chosen Option : 3

Q.77 When a system is undergoing constant volume process, then heat transfer is equal to:

- Ans ☒ 1. work transfer
☒ 2. change in enthalpy
☒ 3. change in internal energy
☒ 4. change in entropy

Question ID : 630680403968
Option 1 ID : 6306801575360
Option 2 ID : 6306801575359
Option 3 ID : 6306801575358
Option 4 ID : 6306801575361
Status : Answered
Chosen Option : 3

Q.78 The process that involves passage of a higher-pressure fluid through a narrow constriction is called:

- Ans ☒ 1. throttling process
☒ 2. polytropic process
☒ 3. free expansion process
☒ 4. hyperbolic process

Question ID : 630680348148
Option 1 ID : 6306801354884
Option 2 ID : 6306801354881
Option 3 ID : 6306801354883
Option 4 ID : 6306801354882
Status : Answered
Chosen Option : 3

Q.79 A sample of an ideal gas is compressed isothermally from volume of 4 l to 2 l. If the initial pressure is 2 atm, what is the final pressure?

- Ans
- ☐ 1. 1 atm
 - ☐ 2. 2 atm
 - ☐ 3. 8 atm
 - ☒ 4. 4 atm

Question ID : 630680503215
Option 1 ID : 6306801966445
Option 2 ID : 6306801966446
Option 3 ID : 6306801966448
Option 4 ID : 6306801966447
Status : Answered
Chosen Option : 4

Q.80 The function of a halide torch is:

- Ans
- ☐ 1. superheating the vapour refrigerant
 - ☐ 2. facilitating better lubrication in the refrigerator
 - ☒ 3. detecting leakage of the refrigerant
 - ☐ 4. defrosting of the cooling coil

Question ID : 630680506409
Option 1 ID : 6306801979150
Option 2 ID : 6306801979151
Option 3 ID : 6306801979152
Option 4 ID : 6306801979149
Status : Answered
Chosen Option : 3

Q.81 The surging phenomenon in a centrifugal compressor occurs when the refrigeration load decreases to below:

- Ans
- ☐ 1. 45% of the rated capacity
 - ☒ 2. 35% of the rated capacity
 - ☐ 3. 55% of the rated capacity
 - ☐ 4. 65% of the rated capacity

Question ID : 630680506395
Option 1 ID : 6306801979094
Option 2 ID : 6306801979093
Option 3 ID : 6306801979095
Option 4 ID : 6306801979096
Status : Answered
Chosen Option : 2

Q.82 The maximum frictional force that comes in play when a body just begins to slide over the surface of another body is called:

- Ans
- ☐ 1. kinetic friction
 - ☐ 2. dynamic friction
 - ☐ 3. static friction
 - ☒ 4. limiting friction

Question ID : 630680394585
Option 1 ID : 6306801538626
Option 2 ID : 6306801538627
Option 3 ID : 6306801538628
Option 4 ID : 6306801538629
Status : Answered
Chosen Option : 3

Q.83 The fluid that becomes less viscous as it is sheared harder is called _____.

- Ans ☒ 1. thixotropic fluid
- ☒ 2. pseudo-plastic fluid
- ☒ 3. Newtonian fluid
- ☒ 4. dilatant fluid

Question ID : 630680508720
 Option 1 ID : 6306801988180
 Option 2 ID : 6306801988182
 Option 3 ID : 6306801988183
 Option 4 ID : 6306801988181
 Status : Answered
 Chosen Option : 2

Q.84 Which of the following claims about the irreversibility of a system is true?

- Ans ☒ 1. Irreversible processes are always spontaneous.
- ☒ 2. Irreversible processes violate the first law of thermodynamics.
- ☒ 3. Irreversible processes violate the zeroth law of thermodynamics.
- ☒ 4. Irreversible processes increase the entropy of the system.

Question ID : 630680503226
 Option 1 ID : 6306801966492
 Option 2 ID : 6306801966489
 Option 3 ID : 6306801966490
 Option 4 ID : 6306801966491
 Status : Answered
 Chosen Option : 4

Q.85 Which of the following is the effect of frictional resistance to the flow of the steam jet over the blade in steam turbines?

- Ans ☒ 1. There is no effect of frictional resistance to the flow of the steam jet over the blade.
- ☒ 2. The velocity of flow at the outlet to the moving blade is equal to that of flow at the inlet to the moving blade.
- ☒ 3. The velocity of flow at the outlet to the moving blade is greater than that of flow at the inlet to the moving blade.
- ☒ 4. The velocity of flow at the outlet to the moving blade is less than that of flow at the inlet to the moving blade.

Question ID : 630680502687
 Option 1 ID : 6306801964333
 Option 2 ID : 6306801964330
 Option 3 ID : 6306801964332
 Option 4 ID : 6306801964331
 Status : Answered
 Chosen Option : 4

Q.86 Which of the following is NOT an example of Newtonian fluid?

- Ans ☒ 1. Water
- ☒ 2. Printer's ink
- ☒ 3. Air
- ☒ 4. Kerosene

Question ID : 630680199007
 Option 1 ID : 630680771755
 Option 2 ID : 630680771758
 Option 3 ID : 630680771756
 Option 4 ID : 630680771757
 Status : Answered
 Chosen Option : 2

Q.87 In which of the following areas is supercharging NOT so important?

- Ans
- ☒ 1. Racing cars
 - ☒ 2. Marine and automotive engines where weight and space are important
 - ☒ 3. Engines working at high altitudes
 - ☒ 4. Domestic bikes

Question ID : 630680502665

Option 1 ID : 6306801964244

Option 2 ID : 6306801964242

Option 3 ID : 6306801964243

Option 4 ID : 6306801964245

Status : Answered

Chosen Option : 4

Q.88 Which of the following is the correct designation of the refrigerant CCl_3F ?

- Ans
- ☒ 1. R 21
 - ☒ 2. R 12
 - ☒ 3. R 11
 - ☒ 4. R 22

Question ID : 630680506454

Option 1 ID : 6306801979331

Option 2 ID : 6306801979330

Option 3 ID : 6306801979329

Option 4 ID : 6306801979332

Status : Answered

Chosen Option : 3

Q.89 The distance of centre of pressure from free surface of liquid is independent of:

- Ans
- ☒ 1. the density of liquid
 - ☒ 2. the moment of inertia
 - ☒ 3. the distance of C.G. from free surface of liquid
 - ☒ 4. the surface area

Question ID : 630680508637

Option 1 ID : 6306801987850

Option 2 ID : 6306801987848

Option 3 ID : 6306801987851

Option 4 ID : 6306801987849

Status : Answered

Chosen Option : 1

Q.90 In a hot working process, metals are deformed _____.

- Ans
- ☒ 1. at any recrystallisation temperature
 - ☒ 2. below their recrystallisation temperature
 - ☒ 3. at their recrystallisation temperature
 - ☒ 4. above their recrystallisation temperature

Question ID : 630680503239

Option 1 ID : 6306801966544

Option 2 ID : 6306801966542

Option 3 ID : 6306801966543

Option 4 ID : 6306801966541

Status : Answered

Chosen Option : 4

Q.91 In a boiler test, the water supplied to a boiler per hour is 2250 kg. The mass of water in the boiler at the end of the one hour is found to be less than that at commencement by 250 kg. The coal burnt per hour is 250 kg. What will be the actual evaporation per kg of the fuel burnt of the boiler?

- Ans**
- ☒ 1. 2500 kg
 - ☒ 2. 250 kg
 - ☒ 3. 10 kg
 - ☒ 4. 8 kg

Question ID : **630680502645**
Option 1 ID : **6306801964165**
Option 2 ID : **6306801964164**
Option 3 ID : **6306801964163**
Option 4 ID : **6306801964162**
Status : **Answered**
Chosen Option : **4**

Q.92 Which of the following does NOT indicate that the refrigeration plant has to be charged with the refrigerant?

- Ans**
- ☒ 1. Reduction in the efficiency of the plant
 - ☒ 2. Difficult to maintain temperature of rooms and holds
 - ☒ 3. Short cycling of compressor
 - ☒ 4. High suction pressure

Question ID : **630680506410**
Option 1 ID : **6306801979156**
Option 2 ID : **6306801979155**
Option 3 ID : **6306801979153**
Option 4 ID : **6306801979154**
Status : **Answered**
Chosen Option : **3**

Q.93 Which of the following statements is true about the fuel feed pump used for diesel engines?

- Ans**
- ☒ 1. The plunger of the fuel feed pump is actuated by the governor.
 - ☒ 2. It is a centrifugal type pump.
 - ☒ 3. When the plunger of the feed pump is lifted upwards, the inlet valve remains open. Therefore, it is not suitable to inject fuel at that time.
 - ☒ 4. It is a spring-loaded, plunger type reciprocating pump.

Question ID : **630680502732**
Option 1 ID : **6306801964511**
Option 2 ID : **6306801964510**
Option 3 ID : **6306801964513**
Option 4 ID : **6306801964512**
Status : **Answered**
Chosen Option : **4**

Q.94 Which of the following factors is taken into consideration when determining the maximum suction height of a centrifugal pump?

- Ans
- ☒ 1. Priming
 - ☒ 2. Cavitation
 - ☐ 3. Manometric efficiency
 - ☐ 4. Mechanical efficiency

Question ID : 630680193035

Option 1 ID : 630680748677

Option 2 ID : 630680748678

Option 3 ID : 630680748679

Option 4 ID : 630680748680

Status : Answered

Chosen Option : 2

Q.95 A closed system undergoes a cycle consisting of two processes. Process 1-2 is an isothermal expansion whereas Process 2-1 is an isentropic compression. According to the first law of thermodynamics, which of the following statements is true for this cycle?

- Ans
- ☒ 1. Heat transfer during Process 2-1 is zero.
 - ☐ 2. Work done during Process 1-2 is zero.
 - ☐ 3. Work done during Process 2-1 is zero.
 - ☐ 4. Heat transfer during Process 1-2 is zero.

Question ID : 630680503235

Option 1 ID : 6306801966528

Option 2 ID : 6306801966525

Option 3 ID : 6306801966526

Option 4 ID : 6306801966527

Status : Answered

Chosen Option : 4

Q.96 Calculate the polar moment of inertia and the maximum torque transmitted by a solid shaft of diameter 100 mm and length 1.0 m. Let the angle of twist be 2° and the modulus of rigidity, $G = 80$ GPa.

- Ans
- ☐ 1. $10.654 \times 10^6 \text{ mm}^4$, 23.525 kN - m
 - ☒ 2. $9.817 \times 10^6 \text{ mm}^4$, 27.409 kN - m
 - ☐ 3. $8.675 \times 10^6 \text{ mm}^4$, 45.500 kN - m
 - ☐ 4. $7.982 \times 10^6 \text{ mm}^4$, 31.542 kN - m

Question ID : 630680205982

Option 1 ID : 630680798734

Option 2 ID : 630680798733

Option 3 ID : 630680798731

Option 4 ID : 630680798732

Status : Not Answered

Chosen Option : --

Q.97 Which of the following statements are correct regarding the characteristics of Entropy?

- 1) It increases when heat is supplied irrespective of the fact whether temperature changes or not.
- 2) It decreases when heat is removed whether temperature changes or not.
- 3) It remains unchanged in all adiabatic frictionless processes.
- 4) It increases if temperature of heat is lowered without work being done as in a throttling process.

- Ans ☒ 1. Only 2, 3 and 4
- ☒ 2. 1, 2, 3 and 4
- ☒ 3. Only 1, 3 and 4
- ☒ 4. Only 1, 2 and 3

Question ID : 630680395246
 Option 1 ID : 6306801541193
 Option 2 ID : 6306801541195
 Option 3 ID : 6306801541194
 Option 4 ID : 6306801541192
 Status : Answered
 Chosen Option : 4

Q.98 The Kaplan turbine is an example of _____ .

- Ans ☒ 1. axial flow turbine
- ☒ 2. mixed-flow turbine
- ☒ 3. tangential flow turbine
- ☒ 4. radial flow turbine

Question ID : 630680193067
 Option 1 ID : 630680748807
 Option 2 ID : 630680748806
 Option 3 ID : 630680748805
 Option 4 ID : 630680748808
 Status : Answered
 Chosen Option : 1

Q.99 Which of the following statements best describes the second law of thermodynamics in relation to cyclic heat and work processes?

- Ans ☒ 1. In a cyclic process, the net heat transfer is always greater than the net work done.
- ☒ 2. In a cyclic process, the net work output can be equal to, greater than or less than the net heat input, depending on the efficiency of the process.
- ☒ 3. In a cyclic process, the net heat transfer is equal to the net work done.
- ☒ 4. In a cyclic process, the net heat transfer is always less than the net work done.

Question ID : 630680503230
 Option 1 ID : 6306801966506
 Option 2 ID : 6306801966508
 Option 3 ID : 6306801966505
 Option 4 ID : 6306801966507
 Status : Answered
 Chosen Option : 1

Q.100 If a point is to be at the free surface of a liquid open to the atmosphere, where the pressure is the atmospheric pressure P_{atm} , then the gauge pressure at a depth 'h' from the free surface becomes:

- Ans
- ☒ 1. $P_{\text{gauge}} = P_{\text{atm}} + \rho gh$
 - ☒ 2. $P_{\text{gauge}} = \rho gh$
 - ☒ 3. $P_{\text{gauge}} = P_{\text{atm}} - \rho gh$
 - ☒ 4. $P_{\text{gauge}} = P_{\text{atm}}$

Question ID : **630680508737**

Option 1 ID : **6306801988249**

Option 2 ID : **6306801988248**

Option 3 ID : **6306801988251**

Option 4 ID : **6306801988250**

Status : **Answered**

Chosen Option : **1**